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Applications of Game Theory in Deep Learning

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Applications of Game Theory in Deep Learning

Springer

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Foreword

In an age of ever-increasing complexity and innovation, the intersection of game theory and deep learning stands as a beacon of intellectual exploration. This book is a testament to the strong bond of two profound disciplines, a fusion that promises to reshape our understanding of decision-making, optimization, and the dynamics of artificial intelligence.

Game theory, originally developed as a framework for understanding strategic interactions, has found new life in the context of deep learning. It provides the tools to model and analyze the complex interactions that occur among agents, whether they are humans, autonomous systems, or algorithms. Deep learning, on the other hand, has ignited a revolution in machine intelligence, enabling computers to learn from data, make predictions, and adapt to changing environments. When these two worlds collide, the result is a profound synergy that opens up a wealth of possibilities.

This volume takes you on a journey through this intriguing landscape, revealing the myriad applications and implications of merging game theory and deep learning. From multi-agent reinforcement learning to the design of robust AI systems, the chapters herein illuminate the transformative potential of this synergy. Whether you are a seasoned researcher, an aspiring practitioner, or simply an intellectual adventurer, this book offers a roadmap to unravel the mysteries of strategic AI and its real-world consequences.

As you delve into the following pages, you will witness how game theory empowers deep learning to tackle the challenges of decision-making in adversarial settings, resource allocation, and strategic cooperation. You will see how it unveils new dimensions of fairness and ethics in AI, and how it is poised to shape the future of industries, from finance to healthcare.

The authors, each a luminary in their respective fields, guide you through these fascinating terrains, sharing their insights, discoveries, and visions. Their work is a testament to the power of interdisciplinary collaboration, forging a path toward a deeper understanding of the potential, as well as the ethical and societal considerations, of AI in a world where strategic interactions abound.

As you embark on this intellectual adventure, keep in mind that the pages you are about to turn represent not just a compendium of knowledge but a gateway to a future where game theory and deep learning will undoubtedly leave an indelible mark. May this journey ignite your curiosity, inspire your own explorations, and lead to a richer comprehension of the applications of game theory in deep learning.

Surat, Gujarat, India Anupam Shukla

Preface

Welcome on a fascinating journey that explores the symbiotic relationship between two of the most significant pillars of artificial intelligence—Game Theory and Deep Learning. The pages that follow will introduce you to a world where strategic thinking and machine learning converge, paving a path for a future that enables intelligent systems to make cognitive decisions.

The ability of game theory to model the strategic interactions of rational agents has long intrigued scholars and policymakers. From economics to biology, game theory has proven to be an invaluable tool for comprehending and predicting outcomes in situations where individuals must balance their own interests with the interests of others. However, in our digital age, where algorithms and autonomous agents are becoming more common, the combination of game theory and deep learning has opened up a new frontier of exploration.

Deep learning, fueled by neural networks, has revolutionized the way computers perceive, evaluate, and acquire knowledge from data. It has accomplished advancements in image and speech recognition, natural language processing, and autonomous control systems. The combination of these two disciplines opens up new and exciting avenues. We observe how artificial agents can think strategically, adapt to ever-shifting environments, and make decisions that are consistent with their goals and the dynamics of their surroundings.

You will embark on a journey of discovery in the following chapters. We have assembled a panel of leading experts and researchers to share their perspectives and findings on the numerous applications of game theory in deep learning. You will learn how this combination can be utilized to improve the robustness and fairness of AI systems, facilitate collaborative behavior among intelligent agents, and address security, privacy, and ethical decision-making challenges.

We invite you to immerse yourself in this book's case studies, methodologies, and real-world applications. We hope that this book will serve as a valuable resource for your understanding and exploration of this flourishing field, whether you are a seasoned professional, a curious student, or an industry practitioner.

We encourage you to think critically about the societal and ethical implications of these technologies as you delve deeper into the intersection of game theory and deep learning. The ability to model, predict, and influence strategic interactions entails a great deal of responsibility. We can collectively shape the future of AI for the betterment of all by understanding its potentials and pitfalls.

We extend our gratitude for joining us on this intellectual journey. This book's contents are a testament to the extraordinary possibilities that emerge when human ingenuity, mathematics, and computational power come together. We hope you find inspiration and insight in these pages, and that this book piques your interest in further exploring the applications of game theory in deep learning.

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Writing a book is a labor of love that often involves the contributions, support, and encouragement of many individuals. We are deeply grateful to those who have been instrumental in the creation of this work, and we would like to express our sincere appreciation.

First and foremost, we want to extend our heartfelt gratitude to our family for their unwavering support and understanding throughout this endeavor. Your encouragement and patience have been my pillars of strength.

We wish to acknowledge the invaluable guidance and motivation provided by Prof. Anupam Shukla, whose wisdom and expertise have been instrumental in shaping the ideas presented in this book. Your insights have been a guiding light on this intellectual journey.

We are also indebted to the numerous researchers and experts who have generously shared their knowledge and perspectives, which have enriched the content of this book. Your contributions are a testament to the collaborative spirit of the academic and professional communities.

We would like to express our appreciation to the team at Springer for their dedication and professionalism in bringing this book to fruition. Your support throughout the publication process has been exemplary.

We are grateful to our colleagues and friends who have provided feedback, encouragement, and a sense of camaraderie during the writing process. Your insights and camaraderie have been a source of inspiration.

Last but not least, we want to thank the readers of this book. Your interest in the subject matter and willingness to explore these ideas are what make the effort of writing a book truly worthwhile.

This book would not have been possible without the collective efforts of all those mentioned and the countless others who have contributed in various ways. Thank you for being a part of this journey.

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Chapter 1
Introduction



In "game theory in deep learning," this book aims to unravel the complex tapestry that interweaves strategic decision-making models with the forefront of deep learning techniques. Our objective is to provide an extensive and insightful exploration, diving deep into both the theoretical foundations and the real-world applications that showcase this intriguing intersection of fields. The journey begins in the introduction, where we lay the groundwork for understanding both game theory and deep learning, highlighting their individual significance and the pivotal role of game theory in enhancing and shaping deep learning algorithms. The structure of the introduction is meticulously designed to guide the reader through a progressive and enlightening journey. Initially, we delve into the essentials of game theory, unravelling its core principles and illustrating how strategic interactions are modeled and analyzed. This sets the stage for understanding various fields' complex scenarios and decision-making processes. Subsequently, we transition into the realm of deep learning. Here, we dissect the fundamental concepts and algorithms that constitute the backbone of deep learning, providing a clear and accessible overview of this dynamic and rapidly evolving area of technology. This section is designed to bring clarity and context to those who are new to the field while offering fresh perspectives to those already familiar with deep learning.

The third section of the introduction bridges these two worlds, illuminating the necessity and impact of applying game theory within deep learning frameworks. We explore how game-theoretic concepts enhance the functionality and efficiency of deep learning models and provide a new lens through which we can interpret and improve these advanced algorithms. Finally, the introduction culminates with an overview of the subsequent chapters, each dedicated to exploring different facets of the synergy between game theory and deep learning. This comprehensive roadmap is designed to orient readers, setting clear expectations for the journey ahead and providing a cohesive narrative thread that ties all the chapters together. We conclude the introduction with a reflection on the overarching themes and the transformative potential of this

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multidisciplinary fusion, setting the stage for a deep and engaging exploration throughout the book.

1.1 Basics of Game Theory

Game theory is a branch of mathematics that studies strategic decision-making. It analyzes situations in which multiple decision-makers, or "players," interact with one another and make choices that affect each other's outcomes. Game theory is used to model a wide variety of situations, including economic markets, political elections, and even evolutionary biology (Ho et al., 2022) to understand the behavior of people in strategic situations. Game theory uses a combination of logic and mathematics to study strategic decision-making. Game theory is used to help businesses—for example, by analyzing whether a new product should be positioned as a luxury or a bargain (Song et al., 2022). A company might use game theory when making strategic decisions about its products and services, as well as those of its competitors. It can also be used to predict the outcome of events involving multiple players.

Games can be classified into two broad categories: cooperative games and noncooperative games (Mirzaei-Nodoushan et al., 2022). Cooperative games are ones in which all the players are acting cooperatively and working toward a common goal. The strategies adopted by each player are intended to achieve that common goal. Examples of cooperative games include checkers, chess, and football. Noncooperative games are ones in which the players have no incentive to work together to achieve a common goal. Players compete to win the competition rather than cooperate to win it. Examples of noncooperative games include poker and rock-paper-scissors. Competitive games are those that are played by two or more players who compete against each other. Winning the game requires having the highest score at the end of the game. A player who adopts a noncooperative strategy (also known as a Nash equilibrium) cannot expect to gain any advantage over his or her opponents by cooperating. In other words, there are no strategies or actions that the other players can adopt to affect the strategy of the player. Players who adopt a cooperative strategy (also known as an NSGNE) have an incentive to cooperate because their combined efforts will allow them to achieve the best possible outcome. This outcome is referred to as a Nash equilibrium because it is an optimal solution that is guaranteed to occur if all the players are rational and strategic in their decision-making.

In game theory, players are often assumed to be rational and to act in their self-interest. They make decisions based on their preferences and the choices available to them. The outcomes of these decisions depend on the choices made by other players (Gimpel et al., 2020). For example, in a game of chess, each player tries to maximize their chances of winning by making the best possible moves given the position of the pieces on the board.

Game theory helps us understand how different players will act in different situations and how these actions will affect the outcome of the game. It is a useful tool for predicting behavior and for finding strategies that will maximize a player's chances of success. Game theory is also used to analyze situations in which the players have conflicting interests, such as in negotiations or the design of economic systems. Game theory is a branch of applied mathematics. It uses mathematical models to analyze strategic situations, in which players have choices that affect other players. Game theory can also be used to model situations such as elections, business interactions, wars, and economic systems. Game theory is a powerful tool that can be used to understand and predict behavior in a wide range of situations, and it continues to be an active area of research today. In the next section, we study deep learning.

1.2 Introduction to Deep Learning

Deep learning is a type of machine learning that involves training artificial neural networks on large datasets (Kelleher, 2019). It is called "deep" learning because the neural networks used in this type of machine learning are composed of many layers, or "depths," of interconnected nodes. These nodes are called neurons because they mimic the activity of neurons in the human brain and work similarly. Each neuron in a neural network receives input from other neurons in the network and then processes this information using an algorithm known as an activation function. The output of the activation function is the next layer of neurons in the neural network, which feeds back into the network via inputs from the previously processed layer. The neural network continues this process by processing the output of the previous layer, resulting in an output that is representative of the input data and a representation of what the neural network has learned.

In deep learning, the neural network is trained to recognize patterns in the data by adjusting the weights and biases of the connections between the nodes. The network is fed large amounts of labeled data and uses this data to learn how to make predictions or perform a specific task. The deep learning model can then be used to make new predictions about new data (Kelleher, 2019). Deep learning is a field of machine learning concerned with algorithms that attempt to model high-level abstractions in data through the use of many layers of nonlinear transformations (Pang et al., 2020). It is part of a broader family of machine learning methods called unsupervised learning, which uses statistical techniques to discover patterns in large datasets. Many modern advances in AI, including machine translation, speech recognition, computer vision, and robotics, are based on neural networks in one form or another. However, most existing deep learning models are complex and require specialized computer hardware to run efficiently. This limits their use in most applications. To train a deep learning model, it is necessary to provide a large amount of

labeled training data. Since this data is very expensive to obtain, researchers have developed several techniques to reduce the cost of training a deep learning model (Hassan et al., 2020).

Deep learning has been successful in a wide range of applications, including image and speech recognition, natural language processing, and even playing games like Chess and Go. It has been used to improve the performance of many different types of systems, from self-driving cars to language translation services. Most modern AI systems are powered by deep learning methods. One of its limitations is that it can only solve problems that can be modeled as a series of linear operations on a collection of inputs (e.g., a set of pixels from a digital image) and a collection of outputs (e.g., a list of words) (Alber et al., 2019). This means that it is only suitable for problems that are easy to describe mathematically (such as classification problems where the outcome can be represented as a vector with one dimension for each type of object). Examples of problems that are not suitable for deep learning include optimization problems, which require models that can learn how to achieve some objective value rather than simply predicting which output will yield it. Another limitation of deep learning is that its performance tends to degrade as it gets further away from its original training set (the set of examples that was used to train the model). This makes it unsuitable for real-world applications where only limited amounts of data are available for training purposes.

One of the key advantages of deep learning is its ability to learn from unstructured data, such as images and audio recordings (Neu et al., 2022). This makes it a powerful tool for tasks that require understanding complex patterns or extracting meaning from large amounts of data. One area where deep learning has made a major impact is image recognition. Recent advances in computer vision have led to the development of technologies such as Google's Deep Dream system (Toğaçar et al., 2021), which uses deep neural networks to process images and generate dreamlike imagery. The system was originally developed for academic purposes and was not intended for production use. However, it has been deployed in applications as diverse as medical research and computer games. Although this technology is still in its early stages, the potential for widespread adoption seems strong. Deep learning is also being used to develop self-driving cars. These vehicles are equipped with LiDAR sensors (Li et al., 2022) that can detect objects in the surrounding environment. The data from the sensors is then analyzed by sophisticated algorithms that can learn to recognize different types of objects and produce a map of the vehicle's surroundings. The main advantage of this approach is that it allows the car's computer to make its own decisions in real time rather than relying solely on input from the driver. Research in this area has yielded promising results, but many challenges remain before self-driving vehicles can be made available for commercial sale. The next section discusses how combining game theory and deep learning can yield interesting results.

1.3 Game Theory in Deep Learning

Game theory can be used in deep learning in a few different ways. One way is to use game theory to model the interactions between multiple agents in a multiagent system (Zhou et al., 2022), such as a group of autonomous vehicles or robots. The agents in the system can be modeled as rational players in a game, and the game theory can be used to predict how they will interact with each other and make decisions. For example, one application of this is for predicting how the agents in a group will respond to certain behaviors of the other agents. Another application is in predicting how different behaviors of an agent will impact the behavior of the agent's neighbors in the network (Monti et al., 2021). Another way that game theory can be used is in the design of neural networks for computer vision tasks (Li et al., 2021). One way to do this is by designing the network to be able to learn to recognize objects from training data which mimics a multiplayer game environment, where each object is represented by one of the players and the other players' actions are used to train the object network to learn to recognize objects correctly.

Another way to use game theory in deep learning is to use it to optimize the performance of a deep learning model (Rajeswaran et al., 2020). For example, in a reinforcement learning setting, game theory can be used to model the interactions between the agent and the environment and to find the optimal policy for the agent to follow. This can be done using techniques such as Nash equilibrium, which is a solution concept in game theory that describes a stable state in which no player can improve their payoff by changing their strategy. Essentially, this means that the agents in this type of setting are locked in their current strategies—they cannot change them to improve their outcomes. This can lead to suboptimal decision-making by the agent and can therefore reduce the overall performance of the algorithm. However, there are techniques available that can overcome this problem by using ideas from game theory, such as dynamic programming or value iteration.

Additionally, game theory can be used to design and evaluate the security and robustness of deep learning models in adversarial settings (Kamhoua et al., 2021), where the model is under attack from malicious actors. Due to the emergent complexity of neural networks and the black-box nature of machine learning, analyzing the security properties of deep networks is a daunting task. There are several approaches to studying the security of deep learning networks using game theory. The first approach is to apply game theory to the problem of adversarial training (Dasgupta & Collins, 2019). Adversarial training entails creating a model that is robust against adversarial examples that are carefully engineered by a malicious attacker to cause the model to make mistakes when used in real-world applications. It is argued that game-theoretic approaches can be used to identify potential weaknesses in the structure and training procedure of a deep neural network that could lead to susceptibility to adversarial attacks. In the second approach, game theory is used to analyze the security properties of classifiers and regression models used in adversarial settings (Chivukula, 2020). A classifier or regression model is trained to classify or predict target variables that an adversary wishes to exploit for malicious purposes. The adversarial examples are defined as pairs of input data and

classification outputs; the goal is to identify regions of the input space that are classified differently than they should be by the classifier. The goal is to identify regions of the input space that are classified differently than they should be by the classifier, intending to exploit those regions to trick the classifier into misclassifying the data as the target variable. Game-theoretic approaches can then be used to analyze the vulnerability of the classifier to adversarial attacks (Pal & Vidal, 2020).

A third approach to studying the security of deep learning networks involves combining game theory and deep learning techniques to analyze adversarial attacks and defenses (Hossain et al., 2022). This approach uses reinforcement learning to train deep neural networks to automatically defend a system from attack, which improves the speed and accuracy of defense systems while reducing the complexity of the underlying algorithms.

In summary, game theory can be used in deep learning to model the interactions between agents in a multiagent system, optimize the performance of deep learning models, and design and evaluate the security and robustness of deep learning models in adversarial settings. Games are one of the fundamental tools for modeling social interactions and decision-making in a wide range of disciplines including economics, psychology, computer science, philosophy, and biology. A common feature of games is that they allow human players to engage in strategic interactions with each other to maximize their performance or that of their opponents. The game-theoretic approach to machine learning aims to enable machines to learn strategies from observed data and use the learned strategies to solve tasks in a fully autonomous manner. Thus, game theory can be used to design algorithms for learning and decision-making in a wide variety of domains such as robotics, artificial intelligence, data mining, econometrics, and finance. In the present work, we explain cooperative game theory, noncooperative game theory, deep learning algorithms, applications of game theory in deep learning, and case studies in the upcoming chapters. In the following subsection, we provide an overview of the chapters presented in the book.

1.4 Chapter Overview

After the introduction chapter, we aim to present the idea of cooperative game theory, Chap. 2: The Dynamics of How Individuals or Groups Work Together to Achieve Common Goals. This chapter lays the foundational concepts and models, illustrating how cooperation can lead to mutually beneficial outcomes. We explain this chapter in the following subsection.

1.5 Cooperative Game Theory

Chapter 2 of “game theory in deep learning” is dedicated to exploring cooperative game theory, an integral aspect of game theory that involves players forming coalitions to achieve mutually beneficial outcomes. This chapter explains how

cooperative game theory operates, its principles, and its diverse applications. Initially, the chapter introduces the concept of game theory, its historical development, and its broad application across various domains such as economics, political science, biology, cybersecurity, and healthcare. It then distinguishes between different types of games, including cooperative and noncooperative games, and discusses the min-max theorem and the concept of zero-sum and nonzero-sum games.

The core focus of the chapter is on defining cooperative game theory. It describes how players in cooperative games form groups or coalitions, aiming for solutions that benefit the group as a whole. This is contrasted with noncooperative games, where players act independently. The chapter uses relatable examples, such as ice cream sharing and voting scenarios, to illustrate how cooperative game theory applies to real-world situations. Important concepts like "coalitional games," "transferable utility," and "stability" in cooperative games are thoroughly explained. The chapter underscores the significance of stability in coalitions, highlighting that a stable coalition indicates members are content with the agreement, reducing uncertainty and conflict.

The "core," a fundamental solution concept in cooperative game theory, is discussed in detail. The core represents a set of stable and feasible outcomes in a coalition game, ensuring no subgroup or individual can gain more by forming their own coalition. This section delves into the mathematical underpinnings of the core and its implications for resource allocation and fairness. Further, the chapter examines "nontransferable utility" and its relevance in cooperative games where direct exchange or utility transfer among players is not feasible, emphasizing the complexities in resource allocation and the importance of fair distribution in such settings. The chapter also introduces the "Shapley value," a method for fair value distribution based on players' marginal contributions to the game. This concept is vital in assessing contributions in collaborative environments where the overall success is a result of the combined efforts of all participants. Additionally, the chapter explores the concept of "dominant strategies," illustrating how players choose strategies that provide the most favorable outcomes irrespective of others' actions and the role of Nash equilibria in optimizing payoffs in multiplayer systems.

Lastly, Nash equilibrium is discussed as a situation where each player's strategy is optimal, given the strategies chosen by other players. The chapter uses practical examples, like the "monkey climb" game, to demonstrate the application and significance of Nash equilibrium in real-world scenarios. This chapter effectively provides a comprehensive view of cooperative game theory, emphasizing its practicality in various fields and highlighting key theoretical concepts for understanding cooperative interactions in game theory. In Chap. 2, as we understand the exploration of cooperative game theory, where the focus has been on the dynamics of collaboration and coalition building, we now turn our attention to a different yet equally fascinating aspect of game theory: noncooperative game theory in Chap. 3. The details in Chap. 3 are in the following subsection.

1.6 Noncooperative Game Theory

The chapter on "Noncooperative Game Theory" in the present book delves into the competitive aspects of game theory where players act independently without forming coalitions or agreements. It contrasts this with cooperative game theory, underscoring the independence and strategic decision-making in competitive environments. Key concepts include Nash equilibrium, minimax strategies, and the absence of binding agreements among players, which leads to a diverse range of strategic interactions in various scenarios like business, politics, and social settings. The chapter further explores various types of noncooperative games, such as the well-known prisoner's dilemma, rock-paper-scissors, and the Friend or Foe game, each illustrating unique aspects of noncooperative strategy. It delves into sequential games like chess and poker, where players' decisions are influenced by previous actions, and discusses the use of decision trees and game trees to analyze these games.

Other notable sections include the examination of games with imperfect information, where uncertainty and hidden knowledge play crucial roles, and games with perfect information, where players have complete knowledge about actions and history. The chapter also addresses advanced concepts like Bayesian games, which incorporate elements of incomplete information, and mechanism design, which explores creating systems to achieve specific goals in multiplayer settings. This chapter provides a comprehensive overview of noncooperative game theory, its applications, and the various strategic considerations involved in such games, making it a valuable resource for understanding competitive strategic interactions in diverse fields. Having delved into the intricate world of game theory, both cooperative and noncooperative, we now start the discussion on the application of game theory in deep learning in Chap. 4. The following subsection explains this idea.

1.7 Application of Game Theory in Deep Learning

The chapter titled "Applications of Game Theory in Deep Neural Networks" in the book covers a wide range of applications, demonstrating the versatility and impact of game theory in enhancing deep learning models and systems. The chapter establishes the foundational concepts of deep learning and neural networks, emphasizing their ability to automatically classify and learn from data without human interference. It then moves into exploring the relationships between neural networks and game theory, using examples like the rock-paper-scissors game to illustrate how game theory concepts can be applied to neural network strategies. Various applications of game theory in deep neural networks are then discussed in detail. These include the following:

1. *Wireless Network Security:* Using game theory and deep learning to protect wireless networks from jamming attacks, with transmitters and jammers engaging in a strategic deception game.

2. *Scene Recognition in Human-Robot Interaction:* Integration of deep neural networks (DNNs) and game theory for advanced scene recognition, crucial in enhancing robot intelligence and interaction with humans.
3. *Adversarial Machine Learning in Cybersecurity*: Implementing game theory to develop strategies for protecting machine learning systems against adversarial attacks, a major concern in cybersecurity.
4. *Mobile Crowdsensing:* Applying game theory in mobile crowdsensing to optimize data collection and improve the resistance of systems against fraudulent data attacks.
5. *Learning Games and Supervised Learning Problems*: Exploring the use of game theory in developing no-regret learning strategies for large-scale games, which can be applied in supervised learning contexts.
6. *Adversarial Attack Strategies in Deep Learning:* Studying various adversarial attacks and defenses in deep learning models, focusing on game-theoretic approaches to understanding and countering these threats.
7. *Nuclear Security:* Using game theory and deep learning to analyze complex data related to nuclear security and terrorism, highlighting the role of game theory in managing nonlinear algorithms for critical societal and business events.
8. *Real-Time Strategy Games:* Application of deep reinforcement learning in real-time strategy games, showcasing the use of advanced AI methods in complex gaming environments.
9. *Green Security Games:* Implementing deep Q-learning in Green Security Games to develop real-time responsive strategies for protecting natural resources against illegal activities.
10. *Multiagent Reinforcement Learning:* Exploring the impact of opponent-learning awareness in multiagent environments, particularly in scenarios involving cooperation and competition.
11. *Multiagent Reinforcement Learning in Traffic Scenarios*: Utilizing deep reinforcement learning combined with game theory for modeling decision-making in multiagent traffic scenarios, emphasizing the strategic interactions among different drivers.
12. *Internet of Vehicles (IoV) and Edge Computing*: Discussing the application of game theory in the IoV for optimizing service offloading and improving the quality of service through advanced computational methods.
13. *AI-Assisted Diagnosis in Dermatology*: Exploring the use of game theory in developing computer-aided diagnostic systems for melanoma detection, employing deep learning models to enhance diagnostic accuracy.
14. *Smart Grid Technology:* Investigating the application of game theory in smart grid systems for enhancing load frequency control and addressing security challenges through artificial neural networks.
15. *Image Restoration Techniques:* Utilizing generative adversarial networks (GANs), inspired by game theory, for image restoration tasks, where the battle between the generative and discriminative models mirrors game-theoretic concepts.
16. *Space Situational Awareness (SSA):* Implementing machine learning and game theory to analyze satellite behavior and enhance the accuracy of behavior classification in space.

17. *Adversarial Attacks and Defenses:* Exploring a game theory framework to understand the interplay between adversarial attacks and defenses in machine learning models.
18. *Resource Allocation in MIMO Networks*: Analyzing the use of game theory in optimizing resource allocation in multiple-input multiple-output (MIMO) networks, utilizing deep reinforcement learning for strategy optimization.
19. *Image Segmentation:* Application of game theory and deep learning in image segmentation, improving feature extraction and generalization capacity.
20. *Autonomous Vehicle Path Planning*: Combining game theory and memory neural networks to enhance decision-making in autonomous vehicles, particularly in complex driving scenarios.
21. *Pricing Strategy for Perishable Food Items:* Implementing game theory and deep learning models to develop effective pricing strategies for perishable goods in supermarkets, addressing environmental and supply chain concerns.

Each of these applications demonstrates the innovative ways game theory principles are being integrated with deep learning technologies to solve. Having journeyed through the theoretical landscapes of game theory in deep learning and its diverse applications, we now arrive at a pivotal segment of our exploration: the case studies. In Chap. 5, we explain case studies. The details of Chap. 5 are described in the following subsection.

1.8 Case Studies and Different Applications

The chapter titled "Case Studies and Different Applications" in the book presents a diverse range of practical applications of game theory in various domains, demonstrating its versatility and depth. The chapter begins by delving into auction theory, a subset of economics and game theory, which studies the mechanisms of auctions and strategic interactions of bidders. It covers the different types of auctions like English, Dutch, and Vickrey auctions and examines how game theory aids in understanding bidder strategies, revenue maximization, and fraud detection. In the realm of pricing, the chapter explores how game theory aids in strategic decision-making by firms in competitive markets, including models like the Bertrand model and analysis of Nash Equilibria in price competition games. It also discusses cooperative pricing games and cartel stability, using examples of organizations to illustrate these concepts.

The application of game theory in generative adversarial networks (GANs) is another focus. It explains how GANs, consisting of a generator and discriminator, can be viewed as a minimax game, a fundamental concept in game theory. This perspective helps in understanding the training dynamics and stabilization of GANs. Furthermore, the chapter touches upon the use of convolutional neural networks (CNNs) in game theory. It illustrates how CNNs can analyze strategic interactions, predict player behavior, and assist in game content generation and strategy recommendations in various game types, including board, card, and video games. In the

context of reinforcement learning, the chapter explains how game theory concepts like Nash equilibria and best responses can analyze the strategic behavior of agents in multiagent reinforcement learning environments. This includes adversarial interactions and the learning process in environments analogous to repeated games.

Additionally, the chapter covers a wide range of other applications, including oligopolies, educational institutions, farming, and computer vision, demonstrating the extensive reach of game theory in modeling strategic interactions and decision-making in diverse fields. Notable mentions include the use of coevolutionary neural population models to simulate strategy evolution in repeated games and game theory in computer vision for strategic interactions in visual scenes. In summary, this chapter provides a comprehensive view of the multifaceted applications of game theory across various domains, showcasing its practicality and the significant insights it offers in understanding and solving complex strategic interactions. The final chapter, Chap. 6, concludes the present book and draws the future research direction. The details of Chap. 6 are presented in the following subsection.

1.9 Conclusion and Future Research Directions

As we draw our exploration to a close in the final chapter of this book, we reflect on the insights gained and the journey undertaken in understanding the intersection of game theory and deep learning. This concluding chapter serves as both a summary of our findings and a forward-looking perspective on the potential future developments in this dynamic field. In this chapter, we synthesize the key concepts, theories, and applications discussed throughout the book, highlighting the most significant contributions and the practical implications of our exploration. We revisit the core ideas that form the backbone of our discussion, drawing connections between the diverse topics covered, from the fundamentals of game theory and deep neural networks to the intricate applications and case studies.

Beyond summarization, this chapter delves into the future of game theory in deep learning. We identify emerging trends, nascent technologies, and unexplored areas that hold promise for further research. This forward-looking section is designed to inspire researchers, practitioners, and enthusiasts to continue exploring, innovating, and contributing to this field. We discuss potential advancements in algorithms, applications in new domains, and the evolution of current methodologies. Moreover, we address the challenges and open questions that remain in the integration of game theory and deep learning. These reflections not only underscore the complexities and nuances of this field but also serve as a call to action for the research community to address these challenges and further advance our understanding.

In essence, the final chapter is crafted to leave readers with a comprehensive understanding of where we stand in the present and a clear vision of the exciting possibilities that lie ahead. It's an invitation to ponder, participate, and propel the future of game theory in deep learning.

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Chapter 2
Cooperative Game Theory



Cooperative game theory is an important area of game theory. In cooperative games, players form coalitions and work together to achieve certain goals. The players basically form groups and then they distribute the amount between them. In cooperative games, players distribute their payoffs. It is called as "coalition game." Examples of cooperative games include situations where players collaborate in business ventures, share resources, negotiate contracts, or work together to achieve mutual benefits. Solutions for Cooperative games include "the Core" and "Shapley Value". Cooperative games give more emphasis to stability and fairness of solution. This chapters presents a number of coordination games, cooperative games, and theoretical concepts of the cooperative games. The chapter also discusses various real-life applications of cooperative game theory.

2.1 Introduction

Game theory is part of mathematics and computer science branch. It deals with situations or problems that have to be analyzed and for which players have to make decisions. Game theory was developed by John von Neumann and Oskar Morgenstern. The book titled *Theory of Games and Economic Behavior* (1944) by John von Neumann and Oskar Morgenstern is considered to be a starting point in game theory. Game theory is applied in various fields like in economics; it is applied to analyze market structures, auctions, and pricing strategies. In political science, game theory has been applied to solve conflict resolution. It is applied in the study of biology as well. Game theory is used to provide insights into various auctions, such as first-price auctions and Japanese auctions. Game theory helps in voting systems, candidate strategies, and in analyzing the results of campaign policies. Game theory is integrated with psychology to study how individuals make decisions under different factors, such as uncertainty and various physical and sensitive emotional conditions.

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In the healthcare and medicine field, it helps in understanding the evolution of drugs so that the medicines can withstand and handle different pathogens. Game theory is used to study the behavior of financial institutions. In the field of cybersecurity, game theory models are used to understand strategies and plans to deal with cyberattacks by providing mechanisms. Game theory models help in explaining the emergence of trust and cooperation in social interactions. It is also used to examine the management of forests, fisheries, and natural resources and to study pollution and its impact on the environment. Game theory plays a significant role in computer science and artificial intelligence. Game theory continues to find new applications in a wide range of fields. Its decision-making dynamics make it a valuable tool in understanding and predicting outcomes in a wide range of real-world scenarios (Rezek et al., 2008). In the classification of game theory, there are several key types of games that help in analyzing different situations and strategies, including cooperative and noncooperative games, extensive form games, symmetric games, asymmetric games, sequential games, matrix games, simultaneous-move games, zero-sum games, and nonzero-sum games. A zero-sum game which is competitive in nature is a game in which the total utility or payoff gained by one player and other players is zero. Here, one player has a positive outcome, and the other player has a negative outcome. The gain and loss of both players are equal, with opposite signs meaning loss of person 1 = gain of person 2 (e.g., = 50, −50). A classic example of a zero-sum game is chess. In nonzero-sum games, the sum of the outcomes of all the players is not zero which means there is no balancing between the players. An example of a nonzero-sum game is the prisoner's dilemma. Real-world scenarios can often be matched with nonzero-sum games. Cooperative games can also be considered as an example of nonzero games. These are nonzero because of the group formation and formation of cooperative policies. A sequential game is one in which the decisions are not made simultaneously. The decision is represented in a tree where the node represents the choice of player, the arrow tells the action, and leaf nodes are also present.

When resources need to be allocated among a group of players or organizations, cooperative game theory can help determine fair and efficient ways to distribute resources and share costs. In business contexts, cooperative game theory can guide decision-making for joint ventures, mergers, and acquisitions by assessing the benefits and risks of collaboration. In the field of artificial intelligence, cooperative game theory can be applied to study interactions among autonomous agents and robots collaborating on tasks. Noncooperative game theory can be used with auctions, as there is competition involved where bidders have to compete to obtain the product.

2.2 Cooperative Game Theory

It could be defined as a game with the formation of groups to obtain the best solution. It basically works from "coalitions." It is also called the "black box" (Hazra & Anjaria, 2022), as the way to find the solution is abstracted. There are several examples to explain it, some of which are as follows:

Ice Cream Example Let us imagine that there are three people, Alice, Bob, and Christine, and they have Rs. 60, Rs. 40, and Rs. 30, respectively. They need to decide how to buy ice cream. Let there be three ice cream tubs of 50 grams, 70 grams, and 100 grams amounting to Rs. 70, Rs. 90, and Rs. 110, respectively. We assume to solve the problem by cooperative or noncooperative approach. The cooperative approach will make more sense in this example, as we can obtain a kind of agreement between the coalition. Alice and Bob need to coordinate their choices to maximize their combined payoff. This ice cream example showcases how game theory can be applied to analyze decision-making and strategic interactions in various scenarios.

Voting Example Voting games basically focus on how individuals can form a group or coalition to maximize their benefits. Basically, their outcomes depend on each player's contribution. These games are often used to understand voting systems, elections, and other situations where a group of individuals have to collectively make a choice. Voting games can have a wide range of applications, including understanding real-world elections, decision-making in committees, and in fields such as political science, social options, and economics. Let's assume a scenario where we have four parties, namely, E, F, G, and H, and have 400 seats, 250 seats, 170 seats, and 180 seats, respectively. In majority voting, the option with the most votes wins. Basically, the party that gets maximum votes win. There is also a "feasibility condition" where the coalition formed should not exceed the seats by the total number of seats. A majority vote is needed so they have to form the coalition and what benefits each party would get, what will be their contributions, and whether they agree to participate in the process or not. In this example, what share will each party earn or what are the agreements that the parties will decide can all be combined and encapsulated in a single term called cooperative game theory.

2.2.1 Coalitional Games

These games are situations where players cooperate and collaborate and are required to achieve objectives such as resource allocation and bargaining. The players in a coalition game work together to enhance their individual or collective payoffs. Coalitional games have the following conditions:

(a) A transferable utility that is actually dividing the outcome.
(b) If we have a set $\boldsymbol{N}$ and the set is finite.

A subset of set $\boldsymbol{N}$ can be a coalition such as {1,2,4,8}, which can be a subset of the above set where characteristic function is defined by $\boldsymbol{V = 2^N}$.

A transferable utility includes a subset and the characteristic function. We get $\boldsymbol{V}$ (members) that is outcome which will come if they do coalition or as the amount of money or utility that the coalition can divide between its members. The game is denoted by both the value function and the subset from which the solution is obtained.

2.2.2 Stability

The concept of stability in coalitional games refers to the notion of stable outcomes or coalitions that are unlikely to break apart due to the players' incentives for defection. Stability is a central concern in cooperative game theory, as it helps to identify stable cooperation structures and predicts how players will behave in coalition formations. Determining whether a given coalitional game has a stable outcome can be challenging, and not all coalitional games have stable solutions. There are different notions for stability, but the most commonly used are core and stable coalitions (Hazra & Anjaria, 2022).

A coalition is considered as stable if no player or subgroup of player has no motivation to leave the coalition to join another one. This concept ensures that no subgroup of players can benefit by forming their own coalition or joining any other group. The aim is to achieve optimality. When a coalition is stable, it indicates that the members have reached an agreement that is resistant to deviations, or we can say that there is no backing off and everyone is satisfied with what they get. Stability can help resolve conflicts more effectively and reduces uncertainty. Stability can also lead to fairness and equality in the outputs and establishes a robust structure for decision-making. Let us understand this through an ice cream example in which there are three sizes of ice cream with small, medium, and large. Small size is of 550 grams, medium is of 700 grams, and large is of 900 grams; A, B, and C have \$5, \$3, and \$4, respectively, and they cost approximately \$7 for small size, \$9 for medium size, and \$11 for large size, and they form grand coalitions For the outcome, we generally have combination of vectors denoted by "$\boldsymbol{X}$", where $\boldsymbol{X} = (\boldsymbol{X_a}, \boldsymbol{X_b}, \boldsymbol{X_c})$ and $(\boldsymbol{X_a}+\boldsymbol{X_b}+\boldsymbol{X_c}) = \mathbf{900}$, which individual rationality will be $(\boldsymbol{X_a} \geq \boldsymbol{v}(\boldsymbol{A}), \boldsymbol{X_b} \geq \boldsymbol{v}(\boldsymbol{B}), \boldsymbol{X_c} \geq \boldsymbol{v}(\boldsymbol{C})$, as they are nonnegative. If these three players play the game, then the outcome will be

Case 1: When there is equal division of ice cream between three players A, B, and C, (900/3, 900/3, 900/3), we obtain 300 as the value of the outcome and this outcome is "nonstable."

Case 2: Let's assume that A and C make a group without B, we obtain 700 grams of ice cream, and their total money sums up to \$9. We have ($\boldsymbol{X_a}$= 350 ,$\boldsymbol{X_c}$= 350, which is better in terms of value, A leaves B and A makes group with C as it will get more good results, but still its "not stable."

Case 3: Let us suppose that there is unequal division and a coalition is formed. If A and B form a coalition and avoids B and their contribution adds up to \$8 which gets the to buy small size ice cream, ($\boldsymbol{X_a}$ = 350 grams and $\boldsymbol{X_b}$ = 350 grams) and B makes an agreement to A that B will provide A 400 grams and it will end up taking 300 grams of ice cream, it is more profitable than the individual division of the amount of ice cream. Group deviations can be considered as breaking the group or dividing the coalition. The set of outcomes we obtain after A and B make a group without C is {A, B} = {400, 300}. Thus, it is also observed that $\boldsymbol{v}(\boldsymbol{A}) = \mathbf{400}$ and $\boldsymbol{v}(\boldsymbol{B}) = \mathbf{300}$ and C did not form a coalition.

In this way, coalitions can be formed and broken. If a member becomes more influential or gains more bargaining power, they may choose to break away or form a new coalition that better serves their interests. Players may receive better offers from players outside the existing coalition.

2.2.3 Core

The core has important implications for understanding how players can cooperate effectively to achieve maximizing outcomes and it's a solution concept popular in game theory. In other word, it could be considered as a set of all stable outcomes; the solution concept helps to identify feasible and stable outcomes when players form coalitions to achieve common goals. The core generates the particular solution which is the optimal one. The pair of vectors satisfies rationality and feasibility for every coalition, and the total payoff is not less than V(C). The core might be empty, which indicates presence of instability and means that there is no solution that satisfies feasibility. So, other solution concepts such as the Shapley value or nucleolus might help in the game. Mathematically, it is written as Core Core $(\boldsymbol{N,v}) = \{(\boldsymbol{x}_1, \ldots., \boldsymbol{x}_n) \in \boldsymbol{R}_n: \sum_{1}^{n} x, \ \boldsymbol{x}_i = \boldsymbol{V}(\boldsymbol{N}); \boldsymbol{X}_i \in \boldsymbol{C}, \boldsymbol{X}_i \geq \boldsymbol{v}(\boldsymbol{C}) \ \forall \boldsymbol{C} \subseteq \boldsymbol{N}\}$.

Or $G = \{ \sum_{0}^{i} x \in C$, such that $\boldsymbol{X}_i >= \boldsymbol{V}(\boldsymbol{C})$ for all $\boldsymbol{c} \leq \boldsymbol{n}\}$, where C is the grand coalition formed and $\boldsymbol{G}$ is the denotion for the core (Hazra & Anjaria, 2022).

Consider a coalition game with a set of players {1, 2, 3} and a characteristic function v(S) that assigns a value to each coalition S of players.

$$v(\{1\}) = 3, v(\{2\}) = 4, v(\{3\}) = 5, v(\{1,2\}) = 8, v(\{1,3\}) = 8, v(\{2,3\}) = \mathbf{9}, v(\{1,2,3\}) = 12$$

Now, let's check if the grand coalition {1, 2, 3} is in the core which is 12. The core conditions require that no smaller coalition has an incentive to deviate. For example, if we consider the coalition {1, 2}, the worth of this coalition is 7. However, if players 1 and 2 deviate and form a coalition just between themselves, the worth would be $v(\{1\}) + v(\{2\}) = 3 + 4$ which gives 7. Since 7 is greater than 8, which makes them want to collaborate with each other. The case is same with other sets also and due to high yield players won't deviate. If the grand coalition is not blocked by any subset of players, then it is in the core. In this example, the grand coalition {1, 2, 3} is indeed in the core because no smaller coalition has an incentive to deviate from it. But in core the value can be null also.

The core has the following properties:

Feasibility: The total payoff allocated to a coalition does not exceed its value. In other words, for every coalition S, the sum of the payoffs allocated to the players

in S is at most the value of the coalition. The sum of the payoffs assigned to the players in S must be less than or equal to the value of the coalition. Feasibility is important because it reflects the practical limitations of resource allocation within coalitions. It also ensures that the payoff allocation is realistic. Individual Rationality: The payoff allocation must be individually rational, meaning that no player should receive less of their worth.

All coalitional games should have a nonempty core. This means that no payoff allocation satisfies both the feasibility and individual rationality conditions. Also, each player in the core must receive the amount as they can receive individually. In certain cases, core might be unique meaning that there is only one stable outcome.

2.2.4 Epsilon Core

The epsilon core is an extension of the core concept in cooperative game theory. It addresses and takes care of situations where the core of a coalitional game might be empty, meaning there is no feasible and individually rational payoff allocation that satisfies all players. It permits individual players to receive payoffs slightly lower than their actual worth in the game, as long as the deviation (epsilon) is within some degree. It helps to capture stable outcomes in situations where the core is empty.

The instability displayed by various examples and some large games (such as the glove market example) can be somewhat overcome using the notion of the epsilon-approximate core. Given any number o, an allocation x is in the epsilon-approximate-core of the coalitional game $(\mathbf{N}, \boldsymbol{v})$ is given by $\boldsymbol{X_i}$ belong to C such that $[\boldsymbol{X_i} \geq \boldsymbol{v}(\mathbf{C})$—o $|C|]$ $\forall C \subseteq N$ where C is coalition value, N is total number of elements, and X_i is particular outcome of player. When the core is empty, certain approximations acknowledge that no coalition has a strong incentive to deviate from the existing arrangement but epsilon cores say that a few has to deviate but it is not enormous bounded by epsilon core-epsilon actually breaking up and may earn high payoff more than before coalition epsilon is very small infimum can be outside of set also highest no less than all no in sets.

In epsilon cores also, the outcomes are present in the feasibility set. It may also have unique values like the core.

2.2.5 Fairness

It involves treating individuals or groups in a morally right and unbiased manner, ensuring that everyone is given equal opportunities and consideration (Narahari, 2022). The main concern in fairness within coalitional games is how to allocate the value or payoff of the grand coalition (the entire set of players) among the players in a way that is considered as equitable. In many organizations, in joint ventures, the

amount is divided in a ratio and shared between various members. The amount that is surplus is also distributed among various members in their collaboration ratio (Hazra & Anjaria, 2022). Therefore, fairness could be considered as the fair amount every individual should receive after their collaboration is formed. For example, if some individual has donated an amount more than the other members, the individual should earn more in terms of profit.

Let us take an example where N is the number of members, $N = \{1,2\}$.

$v(1) = v(2) = 50$, for both players 1 and 2 and $v(\{1,2\}) = 200$ if they both make group. Let us consider the outcome as player 1 earns 150 and player 2 earns 50, which is unfair as player 1 earns more payoff; however, this solution is stable.

$X_1 + X_2 = v$ (player 1, player 2) = 200, which is the outcome generated from the grand coalition. We know that $X_1 \geq 5$ and $X_2 \geq 5$; however, other outcomes could be more stable, such as considering the given outcome $X = (100, 100)$. Therefore, (100,100) is stable outcome, and at some place, they could be equal for them.

Players have some marginal contributions, which are defined as grand group outcomes—individual values. We obtain $v(\{1,2\}) - v(\{2\}) = 200–50 = 150$, and the difference is the marginal contribution:

$$\text{Marginal contribution of player} = \left(\text{Worth of coalition} - \text{worth of individual player}\right)$$

Marginal contribution of 1 group = 150, Marginal contribution of player 2 = $v(\{1,2\})$-$v(\{1\}) = 200–150 = 50$. The payoff should be proportional to each player's contribution, which is called fairness. The marginal contribution to the game is $v(\{1\}) - v(\{\phi\})$ if player 1 exists in the game and there is no coalition. The marginal collaboration of player 1 is empty. Player 1's contribution is the average marginal collaboration of player 1 in group + the marginal collaboration of player 1. Fairness in game theory is often associated with solution concepts that distribute the payoffs among players in a way that is perceived as equitable or justifiable.

2.2.6 Nontransferable Utility

Nontransferable Utility is used to describe situations where individuals or agents cannot exchange or transfer their utility or preferences directly with each other where as "transferable utility" refers to situations where the value or utility associated with goods can be transfered or shared between individuals. This typically occurs in situations where goods are divisible. This concept is particularly relevant in the context of bargaining, negotiations, and allocation problems. Side payments or negotiations are not allowed in nontransferable utility. For example, in a project where team members have distinct roles and responsibilities, the contributions of each member are not interchangeable, which makes it a case of nontransferable utility. This concept has applications in areas such as economics, negotiation, coalition

formation, and resource allocation. Key characteristics of nontransferable utility include the following:

(i) The value formed by the coalition cannot be divided among the members. Each coalition can bring up to certain members or no members at all, which results in indivisibility.
(ii) Since utilities are not transferable, they cannot be easily compared across different players. This lack of comparability makes it challenging to establish a common measure of value or contribution. Cooperative games with nontransferable utility may have unique solution concepts such as the core, the nucleolus, and the Shapley value.
(iii) The allocation of resources or outcomes in NTU settings can be more complex compared to transferable utility. Here, redistribution between different members is not straightforward and easy.
(iv) Players may have specific roles, skills, or responsibilities that mdake them essential in their roles; thus, sharing of utility is difficult, and nontransferable utility settings are typically studied within the framework of cooperative game theory.

2.2.7 Shapley Value

It was introduced by the economist Lloyd Shapley in the early 1950s. It provides a fair way to distribute the total value generated by the strategies of different players based on their contributions to the game (Narahari, 2022). It provides a way to deal with each and every player in a fairer way. The Shapley value has been widely applied in various fields, including economics, political science, and computer science, to allocate resources, assess contributions, analyze cooperative scenarios, and even solve allocation problems in multiagent systems. According to the Shapley value, the amount that player i is given in a coalitional game is

$$\text{Shapley value} = \varphi_i(V) = \sum_{C \subseteq N-i} \frac{|C|!\left(n-|C|-\mathbf{1}\right)!\left(v\left(C \cup \{i\}\right)-v(C)\right)}{n!}$$

Let's consider an example where three players are A, B, and C. For any single player, $v(\{A\}) = 300$, $v(\{B\}) = 400$, $v(\{C\}) = 500$. For any pair of players, $v(\{A, B\}) = 900$, $v(\{A, C) = 700$, $v(\{B, C\}) = 1000$ and $v(\{A, B, C\}) = 2000$. If we try to calculate the marginal contribution for A, when A has different permutations to collaborate, there are different cases as follows:

Case 1: Considering A's cases

$(A, B) \rightarrow v(\{A, B\}) - v(\{A\}) = 900–300 = 600$
$(A, C) \rightarrow v(\{A, C\}) - v(\{A\}) = 700–300 = 400$
$(A, B, C) \rightarrow v(\{A, B, C\}) - v(\{A, B\}) = 2000–900 = 1100.$

Case 2: Considering B's cases

$(B, A) \rightarrow v(\{A, B\}) - v(\{B\}) = 900–400 = 500$
$(B, C) \rightarrow v(\{B, C\}) - v(\{B\}) = 1000–400 = 600$
$(B, A, C) \rightarrow v(\{A, B, C\}) - v(\{B, C\}) = 2000–1000 = 1000.$

Similarly, for C, different permutations can be considered.

The Shapley value for A is $S(A) = (300 + 600 + 400 + 1100)/4 = 2400/4 = 600$. Similarly, for B, the Shapley value is 625, and for C, it is 625. Therefore, according to the Shapley value, A's fair share of the total value is 600, B's fair share is 625, and C's fair share is also 625. The Shapley value(s) considers how much each player contributes to the value of a coalition when joining it. To calculate, we consider the players or a subset of players who come together to collaborate to make groups where all possible coalitions are considered, ranging from individual players to the full grand coalition containing all players. Then, every possible sequence in which players could join the coalition is considered. With each and every player, value is associated, and this value could be in terms of monetary payoffs, resources, or any other relevant measure. To calculate the Shapley value for each player, the average of their marginal contributions across all possible permutations of player orders is taken.

Properties of Shapley values:

(i) The Shapley value works on the principle of average marginal contribution of a player to all possible coalitions they can join. It considers how the player's presence impacts the payoff of other players as they form different coalitions.
(ii) In many cases, the Shapley value provides a unique and consistent solution for the game. The Shapley value is considered a fair allocation method.
(iii) If a player's contribution does not affect the outcomes of any coalition, they are assigned a Shapley value of zero, which ensures that if they do not cooperate, then they will not receive any benefit, and such players are called null players.
(iv) The solution is efficient, which means that the final worth is equally and fairly distributed among the players.
(v) Calculating the Shapley value may be computationally expensive, especially for large games.
(vi) The Shapley value treats players symmetrically, meaning that if two players have equal or similar contributions, then the same Shapley value will be provided to the players.
(vii) The Shapley value for each player is nonnegative, meaning that no player receives a negative allocation.
(viii) There is another property that is a dummy property, which means that this player will have no impact on the coalition, which means that it has no effect on the total worth generated by that coalition.
(ix) The marginal contribution of a player to a coalition is the increase in value that occurs when that player joins the coalition compared to when they are not part of it, such as what they contribute to the group.

Other solutions include those that are similar to shapely values, provide alternatives to shapely values, and handle fairness and stability. Some of them are that the nucleolus provides a solution to cooperative game theory and is also a well-known concept, such as the Shapley value, which provides a fair and stable way to allocate the value or payoff of a coalitional game among the players. It was introduced by Lloyd Shapley and David Gale in 1967 as an alternative to the Shapley value and other solution concepts. It minimizes the excess value that any player claims. It works on the "idea of excess." The excess of a player is the difference between the value they contribute to the coalition and their minimum requirement of participation. It ensures that the payoff allocation is as fair as possible in the sense that the maximum excess among players is minimized. The nucleolus provides an interesting alternative to other solution concepts, such as the Shapley value or the core, and can be particularly useful in situations where players have different minimum requirements or expectations for joining coalitions. One other could be the kernel. The kernel is a subset of the core. Both the core and the kernel deal with the stability and fairness of payoff allocations in coalitional games. It is a subset of the core that provides more restrictions on feasibility. The kernel, unlike the Shapley value, is applicable to games that have nonempty cores.

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Chapter 3
Noncooperative Game Theory



In *game theory*, a noncooperative game is a game in which there is competition between the members, but they do not form groups. In contrast to cooperative game theory, players cannot form coalitions, and they cannot also make agreements. Noncooperative game theory is used in the analysis of various real-world problems, including business competition, auctions, pricing decisions, international relations, and many other strategic interactions. The solutions includes the concept of *Nash equilibrium* and strategies such as the *minimax* mixed strategy which is given by *John von Neumann*. The key difference is that players are not bound to follow any rules. There is no compulsion for them to form groups and share the input and output terms. Every player is independent of the other. This is termed as "independence." The term noncooperative game theory was first used in 1951 by John Nash in an article in the journal *Annals of Mathematics.* Noncooperative game theory includes the number of players, objective function, actions and constraints imposed on the players, and outcome of a probabilistic event.

3.1 Comparing Cooperative and Noncooperative Theory and Their Strategies

John Nash made a statement that explains the difference between the two game strategies. The statement is as follows: "This (cooperative game) theory is based on an analysis of the interrelationships of the various coalitions that can be formed by the players of the game. Our (noncooperative game) theory, in contradistinction, is based on the absence of coalitions in that it is assumed that each participant acts independently, without collaboration or communication with any of the others." There is no binding agreement, and players have to guess or predict the opponent's output. Noncooperative game theory is concerned with understanding strategic interactions in competitive environments without formal agreements, while

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cooperative game theory focuses on the study of how players can work together and distribute the gains from cooperation. Noncooperative game theory involves less inclusion than cooperative game theory. In noncooperative game theory, the concept of Nash equilibrium is used. In cooperative games, the concept of a stable solution is generally more complex. One common solution concept is the core, and the other is the Shapley value. In noncooperative games, players make decisions without any direct communication or cooperation with each other. Communication and cooperation between players are essential in cooperative game theory. Classic examples of cooperative games include the bargaining problem, cooperative coalition formation, and the assignment problem, etc. Examples of noncooperative games include the prisoner's dilemma, Cournot duopoly, etc.

3.2 Nash Equilibrium

A Nash equilibrium in game theory is an idea that describes a situation in which each player's strategy is the best possible choice considering the strategies have been already selected by all other players (Goeree et al., 2002). It is an outcome where neither player has an objective to swap his/her behavior, given the actions of other players. It simply puts in a state where no player can improve their position by changing their strategy while others keep their strategies unchanged. All games do not have a Nash equilibrium, and some may have multiple equilibria. Nash equilibrium provides a mutually optimal solution, meaning that no player can increase or decrease their playoff, assuming that there is no change in players' actions. It is a stability concept and is a regret-free concept, as players are "best responders" to each other's strategies. It does not necessarily lead to the best overall outcome for all players, and it may not capture all aspects of real-world behavior. It has various applications in the fields of economics, politics, biological experiments, etc. Nash equilibrium helps to analyze bidding strategies in auctions. In evolutionary game theory, Nash equilibrium models are used to study the evolution of behaviors and strategies in populations of organisms. Nash equilibrium has applications in political science to study strategic voting behavior in elections and the formation of political coalitions. Companies often use game theory, including Nash equilibrium, to make strategic decisions about pricing, advertising, and product differentiation. It helps predict pricing strategies and market shares.

3.3 Mixed Strategies

In this case, both players randomize their choices with specific probabilities. Mixed strategy equilibria are particularly relevant in games where there is not a clear dominant strategy for any player. In a mixed strategy equilibrium, players assign probabilities to their available strategies. For each player, the expected value or expected

payoff is calculated by multiplying the probabilities and payoffs. The mixed strategy Nash equilibrium captures uncertainty and unpredictability in decision-making. It is a way to find a fair and balanced way to play a game when both players have different options and preferences. The player aims to maximize their expected payoff. In identifying Nash equilibrium there is a series of steps which are needed for optimal combination of strategies of all the players:

1. Recognizing the players, their available strategies, and the associated payoff matrix.
2. These probabilities must sum to 1 for each player along with the assurance provided that no player can increase their expected payoff by moving away from their chosen strategies.
3. Iterating through a process and estimation of the probabilities that each player is trying to allocate to their strategies which would maximize their expected payoff.

Let us consider the most popular game, Battle of the Sexes, which depicts mixed strategy Nash equilibrium.

Battle of the Sexes is a classic game-theoretic scenario that explores the dynamics of coordination and conflicting interests between two players. There are two players who have different preferences in which they want to become engaged. Along with cooperation, there is also competition. There are two choices: either go to fight match or go to ballet. The man prefers to go to fight. The woman wants to go to ballet, and they prefer being together. A higher number indicates a more preferred outcome. The game provides a mixed strategy solution and no pure strategy solution. One player will have a higher payoff in both cases of Nash equilibrium solution. If both players prefer Opera or both players go for a match, they achieve the highest combined payoff (200, 100). But if male chooses match and female chooses Opera, male receives a payoff of 200, and female receives a payoff of 100.

Table 3.1 explains the Nash equilibrium concept by a popular example of a Battle of Sexes game.

We try to assume a case where the man decides to engage in a fight with the probability of q and going for opera with 1 − q probability. Similarly, women choose to participate in a fight with p probability and go for opera with 1 − p probability. To obtain the expected value, multiplication of probabilities with payoff is performed. For men, the payoff for going to opera is 100, and the payoff for going to match is 200. For women, going to opera is 200 payoff, and going to match is 100 payoff. For (fight, opera), the multiplication of probabilities will yield $q * (1 - p)$ and so on. By multiplying the respective payoffs and probabilities, such as the man's expected value for attending a match being q * 200 and for attending an opera being 100 * (1-q), and subsequently solving for q in both equations, we get q = 1/3.

Table 3.1 Payoff matrix of Battle of Sexes game

	Player 2 (woman)		
Player 1 (man)		Fight (p)	Opera (1 − p)
	Fight (q)	(200,100)	(0,0)
	Opera (1 − q)	(0,0)	(100,200)

In the case of women, opting for a fight is associated with a reward of 100 times p, while choosing to go shopping yields a reward of 200 times (1 − p). On solving for p, the value of p becomes 2/3. 100*p= 200*(1 − p), which gives 2/3.

For woman, if p>2/3, then shopping is chosen, but if p<2/3, match is chosen. For man if the value of q>1/3, then he is going for boxing, but if q<1/3, then he attempts to go to shopping.

There can be multiple Nash equilibria or a situation where no Nash equilibria are present. However, Nash equilibrium cannot guarantee optimal solutions; rather, it could generate suboptimal solutions.

There are various examples of games in noncooperative game theory:

Spotlight Game The "Spotlight game" is a popular example in game theory that illustrates the concept of mixed strategy Nash equilibrium. There are two players, namely, Player 1 and Player 2, and there are two conditions: whether to go or to stop; that is, they can either stay or move, which results in four outcomes that are discussed in the form of cases below. In the payoff matrix which is being generated, the first value in each cell represents the payoff for Player 1, and the second value shows the payoff for Player 2. If both players choose to stop, they both have a higher chance of spotting each other and receive a negative payoff. If both players decide to choose "Go" then, this signifies a scenario where they both have a lower chance of spotting each other and will recieve a negative payoff.

Case 1: Player 1 and Player 2 enter the collision state.
Case 2: Player 1 goes and Player 2 stops.
Case 3: Player 2 stops and Player 1 goes.
Case 4: Both players stop.

The second and third cases are Nash equilibrium cases where the outcomes would not be negative and the first and fourth cases lead to negative results. Nash equilibrium occurs when both players randomly choose between "stay" and "move."

Table 3.2 explains the Nash equilibrium concept by a popular example of Spotlight game.

Stag Hunt Game The "stag hunt" game is generally used to explain "Nash equilibrium" concept. In the stag hunt game, there are two Nash equilibria. It is a scenario where two individuals must decide whether to collaborate or act independently, with the goal of achieving the best possible outcome. There are two hunters who have two options: either hunting a stag or hunting a rabbit, which is smaller and easier to catch than a stag. Hunting a hare can be done individually with less efforts but would generate a lower payoff, whereas hunting a stag requires more efforts and

Table 3.2 Payoff matrix of Spotlight game

	Player 2		
Player 1		Go	Stop
	Go	(−6, −6)	(1,0)
	Stop	(0,1)	(−2, −2)

mutual coordination between the players, as it provides a higher payoff. When both players choose to hunt stag again, the solution is in Nash equilibrium. If one player wants to hunt a stag but the other still hunts hare, they will result in a lower payoff, so they decide to hunt rabbit, and thus, the solution is in Nash equilibrium. If one player chooses to hunt stag and the other chooses to hunt hare, the player who hunted the rabbit receives 20, while the other receives 0. We get to know that (10,10) is in Nash equilibrium and (20,0), (0,20) is not in Nash equilibrium. If both players decide to hunt stag, they each receive a payoff of 50. Therefore, if X hunts rabbit and Y hunts stag, X earns 20, but Y will want to change the plan and would want to prefer hunt of rabbit to get 10. In business, companies often face decisions about whether to collaborate with competitors, where to collaborate, and where they basically follow the hunting stag technique while acting alone following the rabbit's plan. Conservation efforts can be seen as a stag hunt scenario. Individuals or nations may need to decide whether to address global environmental issues such as climate change or resource depletion (hunting a stag) or pursue short-term interests that could harm the environment (hunting a hare).

Table 3.3 explains the Nash equilibrium concept by a popular example stag rabbit hunt game.

Rock Paper Scissors Rock-paper-scissors is a classic hand game played in different countries that provides simple and easy examples in game theory and decision-making discussions. It is usually played between two people who simultaneously form one of three shapes with an outstretched hand. The three actions are "rock," "paper," and "scissors." Suppose that if Player 1 chooses scissors and Player 2 chooses stone, then Player 1 loses and Player 2 wins. If Player 1 chooses scissors and Player 2 chooses paper, then Player 1 wins. A few cases have been discussed below:

- Case 1: (paper, scissors) player with scissors will win.
- Case 2: (paper, rock) player with paper will win.
- Case 3: (rock, scissors) player with rock will win.

There are three outcomes: win, loose, and tie, but for tie, there is no collaboration, so they may restart the match. Rock-paper-scissors is an essential game in game theory because it is a noncooperative two-player game with a pure strategy Nash equilibrium. A pure strategy Nash equilibrium is a way in which no player can deviate from their outcome by changing their strategy. There is also no concept of collaboration between teams or players. Let us go through a simple example of a rock-paper-scissors game between two players, Alice and Christine. In round 1, Alice chooses rock, while Christine chooses scissors. Result comes out to be Alice wins because "rock" crushes "scissors." In round 2, Christine chooses rock and

Table 3.3 Payoff matrix of stag hunt game

Player X/Player Y	Y hunts stag	Y hunts rabbit
X hunts stag	50,50	0, 20
X hunts rabbit	20,0	10,10

Alice chooses scissors. Result is Christine wins because "rock" crushes "scissors." Similarly, the game goes on for multiple rounds until there is a clear winner. Rock-paper-scissors can be utilized as a simple and unbiased method for generating random numbers, particularly when a truly random source is not available. In various situations, where parties or various organizations have conflicts and cannot agree on a decision, rock-paper-scissors can be used as a way to resolve disputes.

Prisoner's Dilemma The game provides an example of a situation in which the two players have two choices: either cooperate or betray each other. The prisoner's dilemma was first introduced as a concept in game theory by mathematicians Merrill M. Flood and Melvin Dresher in 1950. The prisoner's dilemma has gained significant attention and popularity in the field of game theory because of its ability to model situations of cooperation and conflict. It is used to study various economic, social, and political scenarios, arms races, business competition, environmental issues, and international relations. It is an important tool in decision-making.

Here, there are two prisoners and they can't communicate with each other and police does not have enough evidence to send them into jail so they make a plan. They provide them with a set of rules that they have to choose, and in this way, the payoff matrix is formed.

The possible outcomes are as follows:

- Case 1: If A and B both betray each other, then, they will be given to be 2 years in prison.
- Case 2: If A backstabs B but B does not speak up, then, A will be removed from prison and B will be in prison for 10 years.
- Case 3: If A remains silent but B backstabs A, then, A will have to be in prison for 10 years and B will be set free.
- Case 4: If A and B both do not open up, then, they will each attend the lesser charge of 4 years in prison.

If both prisoners remain silent and they cooperate, neither of them will be convicted, but they will each receive a moderate sentence for the lesser charge. If one prisoner betrays the other, the betrayer will be set free, while the other prisoner will receive an enormous amount of punishment. However, if both of them confess, then they will have to stay in jail for fewer years. In the payoff matrix, the first number represents the payoff to prisoner A, and the second number represents the payoff to prisoner B. Both players have a dominant strategy that means the best choice for them individually, regardless of the other. If both players follow their dominant strategy, which is to betray each other, they will receive reduced sentences. If both of them cooperate, they would receive a moderate sentence of 1 year each. From an individual perspective, each prisoner has a better choice in betraying the other, as this could lead to anyone of them being set free. However, if both prisoners betray each other, then, they both end up with longer sentences than being silent. The prisoner's dilemma serves as a powerful model to understand situations of conflict and cooperation and is widely used to analyze various real-life scenarios, such as

Table 3.4 Payoff matrix of prisoner's dilemma game

Prisoners A/B	Prisoner B is silent	Prisoner B speaks up
Prisoner A stays silent	Both serve 2 years	A: 10 years, B will have 0 years
Prisoner A speaks up	A: 0 years, B will have 10 years	Both serve 4 years

Table 3.5 Payoff matrix of friend and foe game

Player A/Player B	Friend	Foe
Friend	2, 2	0, 5
Foe	5, 0	1, 1

business competition, environmental issues, and negotiations. In business, the prisoner's dilemma can be used to understand and analyze decisions involving partnerships between companies, competitive strategies, and negotiation tactics.

Table 3.4 shows the moves of players A and B along with their incentives.

Friend or Foe The "friend and foe" game is inspired by the famous prisoner's dilemma, where individuals have the choice to either cooperate (be a friend) or to be away (be a foe) with their partners. It was played in different rounds. *It* was a game show that aired from 2002 to 2003 on the *Game Show Network* in the USA. Each player is given two choices: "friend" or "foe." In this game if both players choose "friend," then it would lead to moderate rewards. But if one chooses friend and other chooses "foe," then that would lead to higher reward to the former. Betrayal by one player ("foe") earns a higher reward if the other remains loyal ("friend"), but if both betray, rewards are lower than mutual cooperation. The game helps researchers to study trust and cooperation patterns among individuals and companies. This is quite similar to the prisoner's dilemma. It is highly used in the areas of economics and also used in behavioral studies and the field of decision-making and psychology. It has the ability to make decisions in a competitive environment. Understanding cooperation and competition is important in organizational settings. The game helps in collaborations between organizations and assures them regarding their outcomes.

Table 3.5 shows the moves of players A and B along with their points and penalties.

Matching Pennies Matching pennies is a two-player game that is a zero-sum game and an example of simultaneous games (Narahari, 2012). Matching pennies is a simple game, and its applications extend to a wide range of fields, providing decision-making. Matching pennies is also used in intrusion detection systems in the study of evolution theory. Matching pennies can be used in cryptographic protocols. For example, if both players randomly choose "heads" or "tails," the expected payoff for each player will be the same. Let us move through an example; suppose there are two players and each player has a penny with two sides, one marked as "heads" (H) and the other as "tails" (T), both have to choose their moves simultaneously without knowing each other's move. If both choose head-head or tail-tail, then the opponent pays some pennies to Player 1. If there are different outcomes, that is, there is a head, and in the other, there is a tail, then Player 1 has to give some pennies

Table 3.6 Payoff matrix of matching pennies game

Player A/Player B	Heads	Tails
Heads	2, −2	−2, 2
Tails	−2, 2	2, −2

to the opponent. Suppose there are two players A and B and Player A chooses "heads" (H) and Player B chooses "tails" (T). In this case, Player A receives a payoff of −2, and Player B obtains 2; that is, Player B wins 2 points, and Player A loses 2 points. The payoffs in this scenario are (A: −2, B: 2). Both players are trying to maximize their payoffs. The payoff matrix is shown below, where the first value is earned by Player A and the second value is earned by Player B.

Table 3.6 represents the moves of heads and tails for both the players and the points which they earn are provided in the cells.

3.4 Sequential Game

It is a game where actions of the first player are chosen before the other players choose the action. To find optimal strategy solution concepts such as Nash equilibrium, backward intuition is used. Chess, tic-tac-toe, and poker are some of the basic sequential games where an unnatural and ever-changing environment is built for the game. Here, players make decisions after taking into consideration of the strategies of previous players. The games that are repeated in several rounds are also examples of sequential games. Extensive forms are applied in sequential games. It's used in making business strategies and in making decisions of investment. In finance, various organizations make strategies based on market values and information available over time, which helps them in making financial decisions. Sequential games also help in making policies in the political environment. Sequential games help political parties to consider possible responses and outcomes. In legal institutions, sequential games are also being used where the next move is being made on the information available and evidence. Combinational games are also examples of sequential games. Simultaneous games, on the other hand, are games where players do not have prior knowledge of the moves of other players. Examples of such games are rock-paper-scissors, prisoner's dilemma, the Battle of the Sexes, the Hawk-Dove game, and the Cournot duopoly game. For analyzing such games, the Nash equilibrium concept is used. Simultaneous games are applied in auctions where bidders give their bids without knowing the bids of other participant bidders. In the field of economics, they are used to analyze market interactions and competition among various companies. In cybersecurity, defenders make plans without knowing the moves of attackers and vice versa.

3.5 Decision Trees

In game theory, decision trees are used in the analysis of sequential games. They provide a systematic representation of the moves, actions, and their respective payoffs. The decision tree helps players to understand the consequences of their actions and make rational decisions based on the strategies of others. The tree representation mainly has nodes that provide information for moves, and the succession of events occurs during the game. The nodes are divided into three types: root nodes, internal nodes, and terminal nodes. Basically, the players need to be identified along with their moves and strategies, and for each terminal node, payoffs are assigned. The concept of backtracking starts from the last level and moves backward, choosing the best strategy for each player at each node based on their payoffs. They are used in bargaining, in producing the best and optimal business strategies, and in making political decisions. It can also be used in making military plans considering potential moves of opposition. It can be used to analyze various investment decisions and resource management.

3.6 Game with Imperfect Information

Games with imperfect information are a type of strategic game in which players have incomplete knowledge or hidden knowledge about the actions or states of the game (Hazra & Anjaria, 2022). There is uncertainty present in the game, which leads to complex solutions and decision-making. Games with imperfect information are usually represented using extensive form game in which players generally use strategic probability and reasoning to make decisions. Imperfect games are more complex than perfect games, as they involve uncertainty. Some common examples of games with imperfect information include poker, board games, auctions, and business negotiations. Analyzing security scenarios, such as intrusion detection, network defense, and cybersecurity games, involves dealing with uncertain or hidden information about potential threats and attackers' actions. Auctions with incomplete information can also be considered as an application of incomplete games. Medical diagnosis and treatment planning often require dealing with uncertain medical data and hidden patient information.

3.7 Games with Perfect Information

Perfect information games (Hazra & Anjaria, 2022) have a scenario where players have complete knowledge about actions taken by other players and the game's history. Perfect information games are often represented using extensive form games or decision trees. The presence of uncertainty is negligible in these games. Here,

players can observe all the plans and actions and generally could be played in a fully known environment. These games are deterministic, and there are no random elements or chance involved in the game. Instances of perfect information games have checkers, tic-tac-toe, and chess. Solution concepts used in perfect information games include backward induction and perfect Nash equilibrium.

3.8 Extensive Form Games

The extensive form could be seen as a graphical representation used in game theory. It generates a structured visual way to analyze the sequential nature of a game and to depict the sequence of actions, strategies, moves, and payoffs in a better manner. It solves games through techniques such as Nash equilibrium and backward induction. It is used for games with imperfect information which is generally used for both sequential and simultaneous games. These games are generally represented in the form of game trees. Economic games, such as the Cournot and Stackelberg models, involve firms making sequential decisions. Let us consider a simple sequential game where Player A makes a choice first and then Player B makes the move accordingly. In the game, suppose Player A chooses between H (head) and T (tail). If Player A chooses H, Player B can choose between center and down. If Player A chooses tail, Player B can choose between left and right. This information could be drawn into a tree, which is shown below (Fig. 3.1).

3.9 Game Tree

A game tree can be considered as a graphical way which specifies all the states within the game, the moves and penalties or rewards (Ho et al., 2022). Game trees measure the complexity of a particular game along with the estimation of the

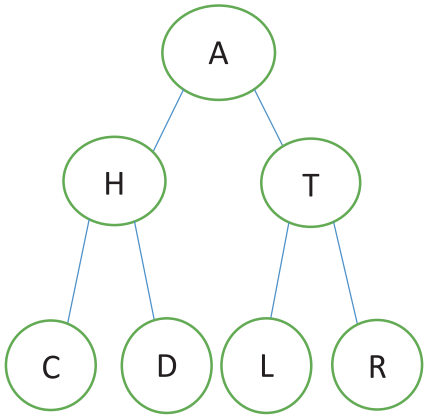


Fig. 3.1 A game tree where A has options of heads and tails and the outcomes by moves

outcomes of moves along with what moves are being taken from a set of all possible moves. They provide real-time scenarios of a game. Games such as tic-tac-toe and chess use game trees. They have certain limitations, such as consuming considerable time. They may or may not provide the best optimal solution. They also check the moves of opponents, and then, from the history and knowledge obtained, they try to make decisions. They need complete knowledge of how to move**.** We also have initial nodes that tell about the initial state of the game. Then, we have edges that represent what moves are to be taken, internal nodes that represent the internal states or the decision that we can take, and, finally, leaf nodes that have final outcomes. For example, the 8-puzzle game has nodes and edges. Nodes could be considered as different combinations of inputs. Edges can be considered different actions, such as moving up or down.

3.10 Search Strategies for Game Trees

There are different search strategies for traversing a graph or a tree. Some of them are listed below.

3.10.1 Breadth-First Search (BFS)

BFS explores all nodes at a given level before going to the next level, meaning it traverses all the nodes in a graph or tree in a breadth-first manner. This allows for a comprehensive exploration of the game's decision-making possibilities. BFS in a game tree is commonly used in artificial intelligence and game development to analyze different scenarios, compute optimal moves, and determine the best strategies for players. BFS can use a very large amount of memory due to the exponential growth in the number of nodes, so it is often applied with heuristics and pruning techniques. It uses queues as the data structure for its implementation. The BFS algorithm ensures the exploration of all the nodes level by level, guaranteeing the visiting of all reachable nodes from the root node. It could be used to analyze all possible moves and their consequences. It also helps in determining the optimal strategy that leads to a desired outcome. BFS also helps in understanding the structure of the game tree, the number of possible paths, and the complexity of the game.

3.10.2 Depth-First Search (DFS)

It uses LIFO (last in, first out) and stack as a data structure. DFS traverses deeper into the tree before considering sibling nodes. It goes deep into levels before backtracking to the previous level. DFS is often used for tasks such as finding a path,

exploring all possibilities, and searching for solutions. It may go into infinite depth and lead to enormous computational effort, so stopping criteria need to be applied. It is used in decision-making scenarios. One disadvantage of DFS is that it may not return optimal path as BFS. DFS is well suited for analyzing sequential games, where players take turns making decisions and can be applied to solve puzzles.

3.11 Min-Max Strategy

Minimax is a decision-making strategy (Ho et al., 2022) in which the objective of one player is to maximize their payoff while the other player aims to minimize it. It is basically used in two-player games and is a kind of zero-sum game. The primary idea behind the minimax approach is to consider the worst-case scenario for themselves and then make moves that minimize the maximum potential loss given the opponent's best response. Basically, it has two players: Player A aims to maximize their own payoff, and Player B aims to minimize A's payoff, effectively trying to maximize their own payoff in a negative sense. This leads to the concept of the "minimax equilibrium," which is a fundamental solution concept in game theory. The minimax algorithm is often used in board games, such as chess and tic-tac-toe. In this scenario, a fully observable environment is needed, and generally, the game uses sequential moves. The algorithm recursively explores the game tree, starting from the current state and considering all possible moves for the current player. When the algorithm reaches a terminal state, it stops and calculates the payoffs which is used to reach the optimal solution, and changes in the payoffs occur during this phase. Minimax rotation has certain limitations. It can be computationally expensive for large game trees and does not work well for games with imperfect information as it requires a fully observable kind of environment. However, this limitation could be solved using alpha-beta pruning.

There are some features of min-max game which are discussed below:

- It is a zero-sum game in which any gain for the first player results in an equal loss for the other player. It involves two players and it's a two-player game.
- It is a perfect game. This means that each player knows the entire game tree, including all possible moves and their corresponding outcomes.
- It works on the Nash equilibrium concept.
- It is alternating game as the game progresses in a sequential manner, with each player reacting to the other player's previous move.
- To determine the optimal strategy in a minimax game, players use backward induction. They start from the terminal nodes (final outcomes) and work their way back to the initial state, making decisions that maximize or minimize the utility at each decision point. Minimax uses a recursive approach to explore the game tree.
- It is a complete algorithm and provides an optimal solution.

- Here, real-world opponents may not play perfectly or may introduce elements of randomness which is a disadvantage (Hazra & Anjaria, 2022).
- The worst-case time complexity of the basic minimax algorithm without any optimizations is exponential (in the depth of the tree), but by alpha-beta pruning, its time complexity is reduced.

3.11.1 Steps in Game Tree

To apply the minimax strategy, the game is typically represented as a game tree, where each node corresponds to a specific game state, and the edges represent possible moves by the players. Firstly, utilities are being assigned. At the leaf nodes of the game tree, payoffs are equal to represent the outcome for each player. In a two-player zero-sum game, the utilities for the players are reverse in signs of each other example −2 and +2. The minimax strategy then applies concept of backward induction. The players maximize and minimize the payoffs while travelling the tree. The minimax strategy helps to identify the optimal strategy for each player to achieve the best possible outcome for their respective goals. In theory, the optimal strategies for both players are found at the Nash equilibrium of the game.

3.12 Strategic Form Games

This is a basic representation of a game using a matrix. Strategic game forms are also called as normal game forms. Each cell in the matrix corresponds to integration of strategies chosen by the players, and the entries represent the payoffs for each player. The number of players is finite along with a finite number of strategies. This allows for a clear and concise depiction of the game's structure. Let us illustrate with a simple example of a strategic form for a two-player game, often referred to as a "prisoner's dilemma": Both players have two strategies: cooperate or do not cooperate. The values in each cell represent the payoffs for Player 1 and Player 2 based on their chosen strategies. For example, if both players choose to cooperate, each receives a payoff of 10.

Table 3.7 represents the prisoner's dilemma, and the payoffs for cooperation and betraying are given in the table.

3.13 Extensive Form Games

These games are useful in situations where players make decisions in a sequence and can choose strategies on previous actions or imperfect information. In other words, it basically depicts sequential games along with incomplete information.

Table 3.7 Payoff matrix of prisoner's dilemma game

Player 1/Player 2	Cooperate	Don't cooperate
Cooperate	10,10	0, 7
Don't cooperate	7, 0	5,5

They are basically used in scenarios such as bargaining and negotiations. These try to represent all the information regarding players, their moves, and their payoffs in the tree form. They can be analyzed using various solution concepts, such as backward induction, which involves working backward from the terminal nodes to get optimal results.

3.14 Dominant Strategies

A dominant strategy is a concept in game theory that represents a player's best outcome or that yields the highest outcome of an action irrespective of the actions made by their opponents. Let us consider a simple example to illustrate the concept of dominant strategy, the prisoner's dilemma, which is a classic game in game theory that involves two players, Player A and Player B, who are given the choice to cooperate or not cooperate with their partners. Let us determine the dominant strategies: If Player B betrays, then Player A's best choice is still to not cooperate (as 1 is greater than 0). Player A's best choice according to the payoffs is to not cooperate (as payoff 6 is more than 2). Therefore, Player A's dominant strategy is "no cooperation" because it always yields a higher payoff. If Player A chooses not to cooperate, then we see in the second row that Player B's best choice is still to not cooperate. But suppose Player B cooperates, then we check in the column of B and consider the two strategies of A. If Player A cooperates, then Player B's best choice is to not cooperate. Player B's dominant strategy is also "don't cooperate."

Table 3.8 represents dominant strategies for both players, which have two choices: either they cooperate or they betray with each other.

3.14.1 Strict Dominance

Strict domination in game theory presents a situation where a particular action for a player results in a strictly greater payoff than another action for every action of the other player. We explain this with the help of "strictly greater than sign." Decisions here are quite easily made, but sometimes, games can have mixed strategies. Suppose we have two players, with Player 1 having two moves A and B and Player 2 having three moves C, D, and E. It is observed that A strictly dominated B for Player 1 when the payoffs of A are always more than those of B in every C, D, and E, which is illustrated in Table 3.9a.

Table 3.8 Payoff matrix of prisoner's dilemma game

Player A/Player B	Cooperate	Don't cooperate
Cooperate	2,2	0, 6
Don't cooperate	6, 0	1,1

Table 3.9a Example of strict domination

Player1/Player 2	C	D	E
A	60	70	80
B	50	60	70

Table 3.9b Example of weak dominance

Player1, Player 2	C	D	E
A	60	70	90
B	60	70	80

Table 3.9c Example of equivalence dominance

Player1, Player 2	C	D	E
A	60	70	80
B	60	70	80

In this case, it is observed that A strictly dominates B for Player 1.

In Table 3.9b, it is observed that A weakly dominates B as 90 > 80.

Equal dominance could refer to a scenario where multiple strategies are equally favorable for a player as they generate same kind of payoff, and therefore, the player is indifferent between those strategies, and they are confronted with a situation of strategic equilibrium. Table 3.9c depicts equivalence of two players with tow moves of Player 1 and three moves of Player 2. A is equivalent to B, which is a case of equivalence dominance.

3.14.2 Weak Dominance

It is a phenomenon where one action could dominate over other action which helps to eliminate less favorable strategies. In other words, it could be considered as a circumstance where a particular strategy for a player leads to at least as high a payoff as any other strategy but not considering what other player chooses. In some games, weakly dominant strategies are not present, so analyzing those games can become more complex which could be said as a disadvantage. Also, there is no guarantee of always higher yield. In this case, equal to is also used along with a greater sign that is "less than equal to." Weak dominance introduces a more lenient form of criterion allowing strategies to coexist.

Table 3.10a depicts weak dominance of middle action of two players with three moves of Player 1 and three moves of Player 2.

In this case, the middle is going to weakly dominate up which could be analyzed. Middle condition is the dominating condition as we can see that Player 2 moves right which results in middle's payoff equal to 60; if movement is in center, then middle is 40; if movement is in left, then it has points.

Table 3.10b depicts weak dominance of center action of two players with three moves of Player 1 and three moves of Player 2.

Player 2 witness's strict dominance of center over left as payoff 40 is greater than payoff 30, and if Player 1 plays middle, then center earns more point than left $60 > 30$ when it goes down $40 > 30$, and if Player 1 plays down, then center earns more points than left.

Player 2 experiences strict domination of center over right, $40 > 20$ in moving left and $60 > 50$ and 60 is superior to 50 in moving right.

Table 3.10c depicts that center strictly dominates right.

Player 1 will go in the middle.

Table 3.10d depicts the middle as the move to obtain a solution. Therefore, the middle appears to be a solution.

Table 3.10a Example of weak dominance

	Left	Center	Right
Up	**20**,20	30,20	30,20
Middle	**20**,30	40,30	60,20
Down	0,0	0,0	0,0

Table 3.10b Example of weak dominance

	Left	Center	Right
Up	20,20	30,20	30,20
Middle	**20,30**	**40,40**	60,20
Down	**25,30**	**25,60**	50,50

Table 3.10c Example of strict dominance

	Left	Center	Right
Up	20,20	30,20	30,20
Middle	20,30	**40,40**	**60,20**
Down	25,30	**25,60**	**50,50**

Table 3.10d Solution to the problem

	Left	Center
Up	20,20	30,20
Middle	20,30	**40,40**
Down	25,30	25,60

3.15 No Dominant Strategy

Let us consider an example of two players, Mike and Bonnie, where both have two actions, going up or going down. First, we need to determine the best responses of Bonnie to Mike's moves. Let us suppose that if Mike goes up in the first column, Bonnie's optimal results are in payoff of 100; when she goes up, it is not zero but 10; when she goes down, indicating up is the best move for her. In column 2, Bonnie's moves are (70-up, 150-down), so down is the best optimal move for Bonnie in a condition when Mike chooses down. There is "no dominant" strategy followed over here. When we consider Mike's moves while traversing rows, they are to go down in the first row with 800 points and to go down in the second row, resulting in 70, which are determined by getting the second value in the cell. Here, Bonnie does not have a dominant move because Bonnie has different moves for different actions of Mike. Mike has a dominant strategy for him and the solution is the down-down strategy (150,70). Therefore, down-down is the Nash equilibrium solution for the problem which is highlighted in the table. In summary, a lack of dominant strategy implies that the best choice for a player depends on the choices made by other players, making the decision-making process more complex and leading to considerations of Nash equilibria.

Table 3.11 explains that no dominant strategies by the Mike-Bonnie game and equilibrium are depicted by highlighted values.

3.16 Bayesian Games

In these games, players have uncertain or incomplete knowledge, and this uncertainty is represented using probability. Players are being assigned with certain probabilities incorporating into account their private information and the beliefs of other players. The solution for such scenarios is Nash equilibrium. These games are not easy to solve, as players have to take complete distribution of strategies and have to take care of their optimal strategies against the opponents. They have applications in the fields of economics, auctions, industries, decision-making, politics, etc. In a Bayesian Nash equilibrium, each player's strategy is optimal given their beliefs about the types of other players, and no player deviate to achieve something better. In traditional game theory, players make decisions based on their own information and the expected actions of other players. The game has various stages, which include the following:

Table 3.11 Example of no dominance

Bonnie, Mike	Up	Down
Up	**(100,** 500**)**	(7, **800**)
Down	(10, 60)	**(150, 70)**

- At the beginning of the game, *nature* chooses the types of each player according to some probability distributions. That is, distribution is determined by nature according to probability distribution.
- Players have *complete information* about their own moves and actions, but they do not have information about other players.
- Players choose their *strategies* based on their private type and beliefs. It takes into account the player's goals, the actions and possible strategies of other players, and the potential outcomes of different choices. Strategies can be classified into several types, such as mixed strategies, dominant strategies, and tit for tat strategies. Then, certain payoffs are being generated with respect to the moves.

3.17 Matrix Games

It is a strategic way of interactions between multiple players where each player has a finite number of strategies to choose from, and the outcome (payoff) of the games is dependent on the strategies (Zhu et al., 2021). The essential elements of the games are players, strategies, and payoffs which they aim to maximize by their decisions which are influenced by the choices made by the other players. The strategic choices that lead to optimal outcomes are often analyzed through equilibrium concepts, the most common one being the Nash equilibrium. A matrix game is defined by a payoff matrix that captures the possible outcomes and payoffs associated with different strategies chosen by the players. The payoff matrix has rows and columns. These rows and columns depict various strategies of different players (Carmon et al. 2020). For example, Player 1 moves can be expressed in rows, and Player 2 moves can be expressed in columns or vice versa. It is a simple way of representing strategic interactions between players. The intersection of rows and columns leads to a cell value that corresponds to the payoff value. Matrix games possess several interesting properties that make them valuable tools for decision-making and analysis in various fields. Some of these properties include the following:

1. Matrix games are represented in a simple tabular format, making them easy to understand and analyze. The matrix shows the possible combinations of decisions made by the players and the corresponding outcomes.
2. They are adhered to noncooperative nature, which means that players independently choose their strategies without direct communication or negotiation.
3. This game involves multiple strategies between two players. Therefore, it's a two-player game.

Let us walk through a simple matrix game example involving two players: Alice and Bob. In this scenario, they are both competing companies launching new products, and they have two choices each: "A" and "B." The matrix will represent their profits based on their combined marketing strategies. In the matrix, the first entry represents Alice's profit, while the second element represents Bob's entry. If both of them choose "A" type, then they will get less profits.

Table 3.12 Example of matrix game

Bob, Alice	A	B
A	$50, $50	$10, $70
B	$70, $10	$30, $30

Table 3.13 Min-max strategy for matrix game

Player A, Player B	First move	Second move	Third move	Row minima
First move	−600	−300	500	−600
Second move	200	0	200	0
Third move	600	−300	−400	−400
Column maxima	600	0	500	Minmax = 0 Max min = 0

Case 1: If both choose "A," both will get $50.
Case 2: If Alice chooses "A" and Bob chooses "B," Alice will have a profit of $10 and Bob will get $70.
Case 3: If Bob chooses "A" and Alice chooses "B," Alice will have a profit of $70 and Bob will have a profit of $10.
Case 4: If both choose "B," then they get $30.

Table 3.12 illustrates matrix games with aggressive and minimal marketing as two strategies and Alice and Bob as players.

Let us consider one more example where there are two players, A and B, and they have three moves to choose from. Let us try to determine the value for this game by using the minimum and maximum strategies.

Table 3.13 illustrates matrix game representation and finding of a solution by using the min-max strategy. By determining the maximum value of the row minima, we obtain max (−600, 0, −400) = 0 as 0 is maximum of all, and selecting the minimum of the maxima values, we obtain (600, 0, 500) =0. We obtain the max of the minima as zero and the minimum of the maxima as zero. Finally, the value of the game is 0 where neither player has any advantage indicating the game is fair. And the strategy for Player A as well as for Player B is also second.

3.18 Repeated Games

Repeated games which are also known as super games are the games where a particular game is played multiple times by the same players. These are represented in extensive forms. Unlike one-shot games where players make decisions only once, repeated games introduce a dynamic element where players' actions and payoffs in one round can influence their choices and outcomes in subsequent rounds. Popular strategies such as tit for tat, the grim trigger strategy, and the triggered strategy are used.

- *Tit for tat* involves cooperation between players in the first round and basically previous moves are seen; if opponent cooperates, then the other player cooperates; but if one defects, the other defects; and this technique is easy to understand and promotes fairness and reciprocity.
- *Triggered strategies* are the ones where Player 1 would collaborate unless and until Player 1 does. Here, players stop cooperating indefinitely. Triggered strategies are particularly effective when dealing with defectors and aim to bring about cooperation.
- *Grim trigger strategies* are strategies in which a player cooperates until the opponent defects, and then the player defects forever, imposing a severe penalty on the opponent.

Repeated games can be classified into finite and infinite games, depending on whether the number of repetitions is finite or infinite.

3.18.1 Finitely Repeated Games

Finite repeated games are a specific class of repeated games where the game is played for a fixed and known number of rounds where players know in advance how many rounds they will play and players receive payoffs from the cumulative decisions made in each round. Also, each round is treated separately, and no history is taken into account while considering the strategy for the present round. One of the most well-known models of finite repeated games is the repeated prisoner's dilemma, which is a classic example of a game with a temptation to defect. Other examples include bidding in an auction with a fixed number of rounds and certain board games such as chess or tic-tac-toe.

3.18.2 Infinitely Repeated Games

These are the games where the sequence of games continues without a predetermined endpoint. In these games, players face a strategic dilemma over an indefinite number of rounds, and their decisions in each round can have long-term consequences for their overall payoff. The two most popular strategies are trigger strategies and the folk theorem. The application of infinite games includes various social and economic contexts, such as business competition, scientific research, and environmental issues.

Let us consider an example where there are two players, and they try to collaborate over some unit of money with an interest rate of r%. If they collaborate, their amount will be $15 each, but if one tries to collaborate and the other tries to leave, the person who leaves will end up in a defect state for the next iteration. The players can collaborate if their amount of collaboration is more than the individual amounts.

Table 3.14 Example of infinitely long repeated game with defect cooperation

Player 1, Player 2	Defect	Cooperate
Defect	(0,**0**)	(**H**,50)
Cooperate	(**50,H**)	(150,150)

Table 3.14 illustrates an example of an infinite repeated game with defective cooperation between two players.

Player 1 assumes defects in the game. So, we assume that Player 2 will not allow Player 1 to collaborate again in the next iteration. Therefore, Player 1 obtains (H, 50) the profit of 50, and then, Player 1 will be in defect state (0,0). Mainly, it can have grim trigger or tit for tat strategy.

3.19 Incentives

Incentives refer to the motivations or rewards that influence the decisions and actions of players in a strategic interaction. They come in various forms and can be positive or negative, depending on how they affect. Players in a game are often rational decision-makers who aim to maximize their own utility or payoff. There are various types of incentives:

Positive Incentives: Positive incentives are rewards or benefits that encourage players to take certain actions. When a player perceives a potential gain or benefit from a particular strategy, they are more likely to choose that option.

Negative Incentives: Negative incentives are deterrents or penalties that discourage players from choosing certain actions. When a player faces potential losses or adverse consequences from a particular strategy, they are less likely to select that option.

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Chapter 4
Applications of Game Theory in Deep Neural Networks



Over the last decade, deep learning has been a hot topic of discussion due to its learning capabilities from data. As a brand-new area of study within machine learning (ML), the deep learning (DL) notion initially emerged in 2006. To understand several applications (Hazra & Anjaria, 2022) of game theory in deep neural networks (DNNs), first let us go through some basic concepts of DL and game theory. Deep learning techniques are a subset of machine learning that is able to classify automatically by learning hierarchical representations in deep architectures. Have you ever wondered how your mobile gallery is automatically organized on the basis of different human faces? This is nothing but the product of DL. Why do we opt for DL in place of ML? In ML, we have to tell machines about the different features that help machines to classify between different species. For example, to classify samples from the mixture of guava and apple, features such as color, size, shape, etc. play an important part. However, in the case of DL, features are picked by a neural network without interference from humans.

4.1 Introduction

Deep learning models are based on DNNs, which are capable of supervised and unsupervised learning with a huge collection of labelled data and back propagation techniques. To understand neural networks (NNs), let us take an example. We have to recognize the letter "A," which is written by three different students. Humans can easily identify letters, but it is possible with DNN, as the handwriting of different students is different. The answer is yes; DNN can classify the letters.

The NN consists of three layers: the input layer, hidden layer, and output layer as defined in Fig. 4.1. In deep learning, the NN is trained to identify letters. They are images of 28×28 pixels. Initially, the image was fed into the input layer. Each pixel

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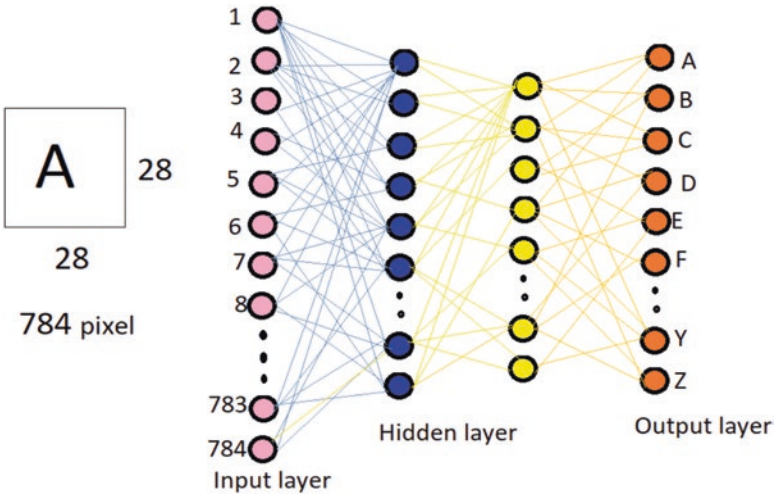


Fig. 4.1 An artificial neural network architecture

of a given image is fed into the input; let us say X_1, X_2, …, X_{784}. Now, it passes through a hidden layer of a channel. Each channel consists of several weights. Hence, they are called weighted channels, which can be represented as W_1, W_2, …, W_{784}. All the neurons of hidden layers are associated with numbers called bias, which can be represented by b_1, b_2, …, b_{784}. Now, we have to find a weighted sum for all input neurons $X_1 \cdot W_1 + X_2 \cdot W_2 + \ldots + X_n \cdot W_n$:

$$\sum_{n}^{i=1} X_i \cdot W_i + b_i$$

After that, bias is added. The activation function, which aids in identifying neurons that need to be triggered, is then passed via the summing function. Until the second-to-last layer, each stimulated neuron transmits information to the next layer. The letter is represented by one neuron in the output layer that is active. Back propagation is a technique for continuously adjusting weights and bias to create well-trained models. There are numerous applications of DL in healthcare, business, agriculture, research, and many more that you cannot even imagine.

Healthcare: Regular health factor analysis, coronary heart disease risk prediction, cancer classification, diagnosis of COVID-19, detection of COVID-19.

Natural language processing: Text summarization, sentiment analysis.

Cybersecurity: Malware detection, suspect detection, network intrusion detection, security incident and fraud analysis.

IoT and security: Smart parking system, air quality prediction, cybersecurity in smart cities.

Smart agriculture: Plant disease detection, soil quality evaluation, smart agriculture IoT system.

Business: Stock trend prediction, financial loan default prediction, power consumption forecasting.

There are also some limitations of DL. Deep learning is the most efficient way to deal with unstructured data. To train a neural network requires considerable data, and processing a huge amount of data is not possible for every machine. Another limitation is that the computational power to train the neural network requires graphical processing units (GPUs). The GPU consists of thousands of cores compared to CPUs. The GPU is more costly compared to the GPU. Another limitation is the training time to train a neural network. It takes days or months. The duration of training is dependent upon amount of data and the number of layers in the network. Now, after understanding the basic concept of networks, let us enter the application part of game theory. Therefore, there are many applications of game theory in deep neural networks. It is critical to comprehend the relation between game theory and deep neural networks.

4.2 Relation of Neural Network to Game Theory

To understand it in a better way, let us consider an example of a game (Van den Nouweland, 2007). We have two players, Joy and Roy. The main components of a game are the players of the game, rules of a game, and output of a game. To understand the behavior of players' strategy, a game payoff matrix is used for representation. Therefore, all of us are aware of a rock-paper-scissors zero-sum game in which players simultaneously choose either rock, paper, or scissors. Let us discuss the basic rule of this game.

Rules:

1. If Joy opts for rock and Roy opts for paper, then Roy will win one point and vice versa.
2. If Joy opts for paper and Roy opts for scissors, then Roy will win one point and vice versa.
3. If Joy opts for rock and Roy opts for scissors, then Joy will win one point and vice versa.
4. If Joy opts for paper and Roy opts for rock, then Joy will win one point and vice versa.
5. If Joy and Roy both opt for the same, then both will get zero points.

Figure 4.2 shows the payoff matrix as per the given game rules. The leftmost point of each cell is the point of Joy, and the rightmost point of each cell is the point of Roy. For example, if Joy opts for paper and Roy opts for rock, then Joy will get 1 point and Roy will get −1 point.

There are two types of games: static games or dynamic games. A game in which complete information such as strategies and the payoff matrix is available for another player is called a static game. Games in which incomplete information is present, unlike static games, are known as dynamic games. The above game is a

JOY \ ROY	rock	paper	scissors
rock	0,0	-1,1	1,-1
paper	1,-1	0,0	-1,1
scissors	-1,1	1,-1	0,0

Fig. 4.2 Payoff matrix of rock-paper-scissor game

JOY \ ROY	rock (l)	paper (m)	scissors (1-l-m)
rock (x)	0,0	-1,1	1,-1
paper (y)	1,-1	0,0	-1,1
scissors (1-x-y)	-1,1	1,-1	0,0

Fig. 4.3 Payoff matrix of rock-paper-scissor game with mixed strategies

static game; as in this game, all players have to play simultaneously. In this game, players are unaware of others' strategies but play according to their assumption. In dynamic games, players make their decision sequentially (e.g., game of chess).

Therefore, in this game, there will be no Nash equilibrium in pure strategies (players choose action for sure), as the player who loses or ties can always switch to another strategy and wins the game. We shall look into mixed strategy. Now let us consider x, y, $1 - x - y$ as the chance of picking rock, paper, and scissors, respectively, for Joy. Similarly, we can say l, m, $1 - l - m$ is the chance of picking rock, paper, and scissors, respectively, for Roy. Each player's set of actions is {rock, paper, scissors} and (p(rock), p(paper), p(scissors)) $\geq$ (0,0,0) and {p(rock) + p(paper) + p(scissors) = 1}.

Mixed strategy responses by Joy, Joy's expected payoff from playing the mixed strategy (x, y, $1 - x - y$) when Roy plays mixed strategy (l, m, $1 - l - m$). According to Fig. 4.3, the payoff amounts to Joy:

$$r = x{\cdot}l{\cdot}0 + x{\cdot}m{\cdot}(-1) + x{\cdot}(1-l-m){\cdot}1 + y{\cdot}l{\cdot}(1) + y{\cdot}m{\cdot}0 + y{\cdot}(1-l-m){\cdot}(-1) + (1-x-y){\cdot}l{\cdot}(-1) + (1-x-y){\cdot}m{\cdot}1 + (1-x-y){\cdot}(1-l-m){\cdot}0$$

$$= -xm + x - xl - xm + yl - y + yl + ym - l + lx + ly + -mx - my$$

$$= -3xm + x + 3yl - y - l + m = x + m - l - y - 3xm + 3yl$$

Our main goal is to maximize the payoff expression. From the expression, we can see that the payoff expression of Joy also depends on Roy's probability (l, m). Similarly, we can find the payoff expression of Roy according to Fig. 4.3:

$$q = l{\cdot}x{\cdot}0 + m{\cdot}x{\cdot}1 + (1-l-m){\cdot}x{\cdot}(-1) + l{\cdot}y{\cdot}(-1) + m{\cdot}y{\cdot}0 + (1-l-m){\cdot}y{\cdot}1 + l{\cdot}(1-x-y){\cdot}1 + m{\cdot}(1-x-y){\cdot}(-1) + (1-l-m){\cdot}(1-x-y){\cdot}0$$

$$= xm - x + xl + xm - yl + y - yl - ym + l - lx - y - m + mx + my$$

$$= 3xm - x - 3yl + y + l - m = y + l - x - m + 3xm - 3yl$$

Similar to the last expression, this expression of the payoff of Roy also depends on Joy's probability (x, y).

As we are aware of the concept of neural networks, in the last section, we discussed the game rock-paper-scissors, in which two human players, Joy and Roy, were involved. Imagine that if the game is played between a human and an agent, how will an agent be able to understand which strategy to choose? Learning is needed to develop such systems. From the last example of game rock-paper scissors, we know that with the help of the payoff matrix and mixed strategy values, it becomes easy for a player to decide which strategy to use. An algorithm can be developed in which we have to perform a simple classification task. First, to understand the algorithm properly, let us take a game with a payoff matrix of size 2*2. In place of players, we will consider neurons who are participating in the game.

Let's consider a game in which there is a race to complete and there are two options: either run or jump. To determine the reward matrix for the payoff function and the mixed strategies, we require a learning algorithm for neural networks based on a theoretical understanding of game theory.

In Fig. 4.4, all are unknown, and to find the values, an algorithm was used. Let us consider a supervised learning classification task that is one dimensional, linearly separable, and simple. Here, we need to classify between two classes: run and jump. For classification, we are considering m objects, and these will be the training set for the learning algorithm. Therefore, m objects of two different classes are represented in the x-y plane, and each object i belongs to X_i [0,1].

Consider red balls as class-1 representing run and blue balls as class-2 representing jump state. There are two points q and q′. q′ is able to correctly classify between two classes, but q misclassifies the two balls as shown in Fig. 4.5. At the beginning of the learning process, q is placed randomly on the x-axis and starts moving to the origin until it reaches q′. Consider a mixed strategy for neuron 1 (p, 1-p) expressing

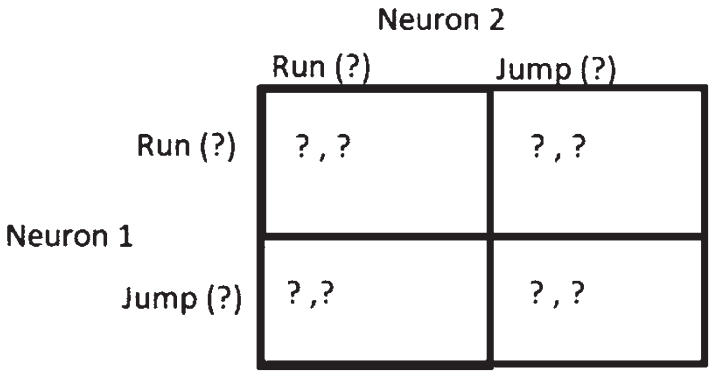


Fig. 4.4 Payoff matrix

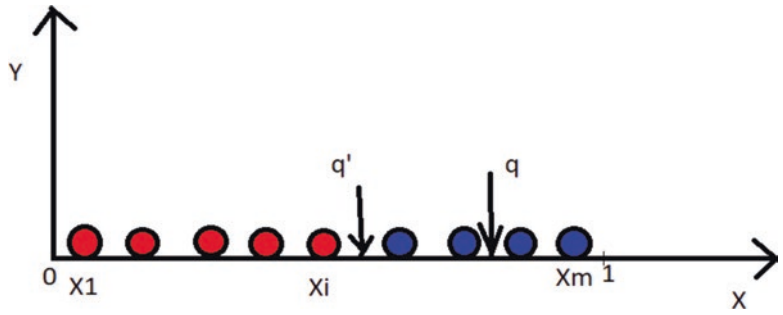


Fig. 4.5 Classification of two classes that are linearly separable and 1D

neuron-1's scepticism toward neuron-2, and neuron-1's objective is to build a model of neuron-2's anticipated behavior.

In Fig. 4.6, f_1, f_2, and f_3 are the payoff functions. Here, f_1 is fixed and represents the payoff function of class-1, which is the run state of neuron-1. f_2 is found by the angle R [0,90°], which is a function of class-2, jumps to the state of neuron-1. As p belongs to [0,1], it is directly proportional to angle R, which belongs to [0,90°]. For example, if p = 0.5, then angle R is equal to 45°. q represents random assignment, and function f_2 is formed. Now, we increase the angle R, and finally, at angle S, classification is done correctly. There are several such steps to reach at point q'. Now, the intersection point of payoff function f_1 and $f_3$is the final solution to our problem. From this position, we are able to find the payoff functions and mixed strategies of neuron-1, which is the goal of our learning algorithm.

Keeping an eye on point x_1, which is located to the left of p' and intersects our necessary payoff functions at f_1 and f_3. At this point, $f_1(x_1) > f_3(x_1)$. Thus, the payoff for the run state is always greater than that for the jump state. Similarly, we can see $f_3(x_3) > f_1(x_3)$, which means that the payoff function of the jump state is greater than that of the run state. As a final result, we can say that f(x) = run for $x < x_{q'}$; otherwise, jump.

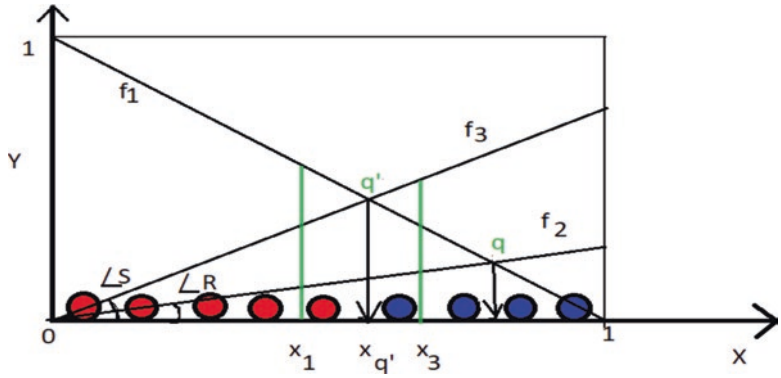


Fig. 4.6 From the perspective of the learning algorithm, neuron-1

4.3 Applications

We have discussed some basics of deep neural networks and the relation between game theory and deep neural networks. Now, we will focus on the applications part.

The authors (Weerasinghe et al., 2018) discussed that using theoretical game theory and a deep learning approach can save wireless networks from jamming attacks. The main contribution is using adversarial (deep) learning approach. The transmitter and jammer aim to deceive one another by purposefully faking data that the opponent uses to build judgments. Therefore, to understand jamming-related problems, the authors used a game theory approach. Let us focus on the problem statement where the transmitter has to form a complete communication network with receivers and the job of the jammer is to get in the way. By increasing the signal-to-interference plus noise ratio at the receiving end, a jammer might disrupt communication. Therefore, instead of using a particular channel for communication over a defined period of time, transmitters can shift to other available channels with some probability. The jammer can also find the probability of the transmitter shifting over that particular time period, which can cause a jammer attack. It can be prevented when the transmitter changes its probability periodically. The transmitter uses a pseudorandom number generator function for the probability distribution, which changes periodically. When the probability of transmitter shifting changes randomly, it becomes difficult for the receiver to communicate with the right channel. The jammer would need to know the probability distribution to be utilized in the upcoming time interval to successfully halt transmission. The transmitter can respond by either increasing the adversarial distortion intensity or by changing generating functions if the jammer is effective in predicting the probability values over time. The jammer also intentionally modifies its jamming patterns to fool the transmitter to believing that it has not been able to figure out the elementary designs.

In paper (Wang et al., 2019), the work suggests a scene recognition model with applications to human-robot interaction that is built on DNN and game theory.

Conventional scene identification systems typically use either low-level features or high-level features. In addition to being clear-cut and easy to use, DNN and game theory also offer the benefits of solid logic and congruence with human intuitive perception. The volume of scene data can also be accurately described by deep learning-based methods. The main components of the deep learning-based scene identification system are front-end local descriptor development, back-end optimization, and loop detection of the created map. The key functions of the front end are feature point extraction, camera position calculation, and local map generation. The preprocessing operation is crucial for the effective processing of the photos. In this article, the author discusses using game theory to complete the task. The two most frequently used ways are the "spatial domain" and the "frequency domain." Game theory is used to handle simultaneous making of decision concerns in a dispute surrounding. Large systems are decoupled into modules by each module, which build classified judgments establish on data relation. To complete the registration, nonzero and competitive games between the two players are played using the example of the two characteristics as two players. In wagering, two participants want to maximize their personal funds or keep the cost to a minimal level, leading to two different types of features known as decouplings. By using rational decision-making, the wagering balanced frame-matching solution is obtained, and this proportion enhances a straightforward desired performance for universality. The author found the game's Nash equilibrium point, which provides the lowest energy level and completes the image preprocessing process, to make use of the EM model to obtain the best result. Currently, the most important technology for increasing a service robot's intelligence is vision-based scenario analysis and identification, which is also the first step in making intelligent robots. Using DNN-assisted scenario identification, it is possible to continue the scene's descriptive analysis and training, the scene's recognizable leak labelling to the technique of recognizing prior periods, the tags made error, and the sample's processing improvements, which creates the new dominant character collection indefinitely. Game theory is at the heart of this paper. Each person in the game possesses a unique set of tactics because the images are thought of as a collection of people. The model's degree of freedom can reach infinity, which helps improve accuracy and registration ability.

In paper (Dasgupta & Collins, 2019), the authors discussed, in cybersecurity tasks, adversarial machine learning: game theoretic approaches. Using the computational framework of game theory, this paper presents methods considered to build a machine learning system resistant upon adversarial attacks. In an example of machine learning termed adversarial learning, two parties known as the learner and adversary make an effort to develop a prediction mechanism for data pertaining to the current problem domain, but with various goals. To accurately categorize or predict the data is the learner's goal when learning the prediction mechanism. The adversary's goal, on the other hand, is to misguide the learner into generating inaccurate predictions about the facts in the future. Adverse learning poses a serious threat to cybersecurity in several areas using machine learning-based classification systems, including automatic spam filters, antivirus programs, and classification formulas, image type in defense and security, medical apps, and many more. Game

theory provides an interesting tool for adversarial learning because it provides a way to mathematically model learners and enemy behavior in terms of defensive and offensive plan and to identify appropriate schemes to demote the loss of learners due to conflicting attacks. This paper discussed the background of game theory and adversarial learning.

The authors (Li et al., 2017) discussed a safe mobile crowd sensing game with deep reinforcement learning. As we all know, our smartphones consist of a number of sensors, including accelerometers, and mobile crowdsensing (MCS) provides location-based services. An MCS server seeks to enlist a few nearby smartphone users to collect sensory data and create an MCS application. Game theory is used to formulate MCS processes to provide auctions and pricing and offer an approach of reputation-based mechanisms to encourage user participation in MCS services. In this game, the server uses a classification algorithm to assess each sensor report's accuracy and determine the winner. To reduce the incentives for smartphone sensing attacks, each user is compensated according to how accurately they use their senses. Users who cheat are not paid anything. To understand the effects of the detecting expenses, each user's participation makes to the precision of the webserver, the quantity of current smartphone owners, and the MCS software in evaluating the sensing; the Stackelberg equilibria (SEs) and safe MCS game are calculated and supplied. Increasing numbers of people are using smartphones to aid with fraudulent detection attacks, and smartphone crowdsensing software is reduced by a larger reward. The overpayments, however, cause oversensing and can occasionally cause network congestion, which reduces the server's usefulness.

A Q-function, or quality function that represents each state-discounted act's extended payout combination, along with the current condition of the preceding recognition of the payment schedule and report performance, is used by the Q-learning-equipped MCS server to determine the payment procedure. The high-dimension curse serves as an example of the sluggish learning rate based on Q-learning MCS system in the presence of a sizable state space. One solution to this problem is the deep reinforcement learning network, which is a 2014 video game deep Q-network (DQN) technique developed by Google DeepMind. More precisely, they suggest an MCS payment approach based on DQN that utilizes a deep convolutional neural network (CNN) to calculate each Q-value payment value and compact the training subspace. By utilizing Q-learning and deep learning, this MCS payment plan expedites the process of determining the optimal payment policy to acquire, so enhancing the behavior of the suspicious system in defense against attacks utilizing faked sensing.

The authors (Schuurmans & Zinkevich, 2016) discussed how the monocyclic neural network with the help of determinable convex gates is used in a game by generating a bijection among the deep learning's critical (or KKT) points and Nash equilibria issue. The researchers of the paper want to find a novel way to reduce guided learning to gaming. The employment of no-regret strategies to resolve large-scale games as efficient training with stochastic techniques for supervised learning problems is an intriguing discovery. Authors begin by thinking about the simpler one-layer learning problem (OLP), allowing us to present the fundamental ideas. Then, deep models will be added. The authors begin by identifying a

straightforward game whose Nash equilibria match the global minimum of the one-layer learning problem (OLP). This fundamental connection creates a link between supervised learning and playing games. According to the OLP, the authors proposed different games: OLG (one-layer learning game), OCP (one-layer constrained learning problem), and OCG (one-layer constrained learning game).

Regret matching (RM) and the normalized exponentiated weight algorithm (EWA), a more straightforward technique from the research on economy and game theory, were two algorithms that the authors took into consideration for learning from expert advice. These algorithms conduct their updates for supervised learning using a random sample of the gradient. The author made a comparison between these and projected stochastic gradient descent (PSGD), which is a clear modification of stochastic gradient descent (SGD) that nonetheless has a similar regret bound. The authors ran experiments on both the MNIST dataset and synthetic data to examine the usefulness of these strategies for supervised learning. This research makes a significant contribution by demonstrating the gamelike nature of the task of training a feedforward neural network with differentiable convex gates. A useful outcome of this reduction is that it recommends novel training strategies for deep models that are motivated by techniques that have recently demonstrated success in trying to solve massively multiplayer online games. Many studies take regret minimization into account to address offline optimization issues. Currently, Adagrad and conventional stochastic gradient descent are two well-liked strategies. The concept of selecting a minimizer from a specific family of functions to simplify the class of losses first appeared in the literature on regret minimizing and has subsequently been expanded upon.

In paper (Ren et al., 2020), the theoretical underpinnings, methods, and applications of adversarial attack strategies are introduced. The machine learning and security industries have both been paying growing attention to adversarial attack and defense approaches, which have emerged as a popular study topic in recent years. The research community has, however, identified a serious risk to the security of the current DL formulas: With the manipulation of benign samples, adversaries can quickly trick DL models without being noticed by humans. Imperceptibility to human hearing and vision changes is sufficient to cause the model to forecast incorrectly with a high degree of confidence. The adversarial sample phenomenon is seen to be a substantial barrier to the widespread use of DL models in industrial settings.

Current adversarial attacks can be divided examining gray-box, black-box, and white-box attacks in accordance with the threat model. The adversaries are presumptively fully aware of the parameters and architecture of their target model in the white-box attack threat model. The target model's architecture is the only thing the adversaries are aware of in the gray-box threat model. The only method the adversaries have in the black-box threat model to produce adversarial samples is query access.

Heuristic and certificated defenses, among others, have recently been proposed as defensive methods for adversarial sample detection/classification. Heuristic defense is a term used to describe a defense mechanism that successfully counters specific attackers despite lacking theoretical precision assurances. The most

effective heuristic defense at the moment is adversarial training, which aims to increase the resilience of the DL model by including adversarial samples in the training phase. Since most heuristic defenses cannot stop adaptive white-box attacks, the community is beginning to concentrate on certificated defenses. A certificated defense is meant to ensure defensive effectiveness in specific circumstances independent of the attack strategy employed by adversaries. The author of this work examines and reviews the adversarial attacks and countermeasures that reflect the most recent developments in this field.

The following are particular adversarial attacks on the other DL models, such as the L-BFGS algorithm, fast gradient sign method, BIM and PGD, momentum iterative attack, distributionally adversarial attack, Carlini and Wagner attack, GAN-based attacks, and many more. The author of this work provides a summary of various typical defenses that have been introduced recently, primarily randomization-based schemes, adversarial training, and denoising techniques, proven defenses, and a few additional fresh defenses.

In paper (Woo, 2019), the overall idea is to provide nuclear security with the help of a nonzero-sum game and deep learning. There have been many attacks by attackers on nuclear power plants, and to ensure security, game theory plays an important role. To quantify the safety of nuclear power facilities, a complicated nonlinear algorithm is employed using the nonzero technique of game theory. Moreover, deep learning is employed in the same areas of data computation as neural networking. There are various ways to manipulate data, including information retrieval and the analysis of event structures, which allow for a deeper investigation of the data. Game theory's nonlinear complex algorithm is used to analyze terrorism-related information because it provides a more accurate description of complex phenomena such as terrorist assaults on nuclear infrastructure. For modeling purposes, it uses a payoff matrix that consists of five different factors of nuclear security and two different factors of nuclear terrorism. Then, with the help of the payoff matrix, a game tree is built, and with the help of the game tree, a payoff graph is made. One of the key attributes is the ability to manage nonlinear algorithms, which require no exact solutions to identify the outcomes of investigations, particularly for vital events in society and business. It has been a high focus to examine human error in addition to safety and security.

In paper (Arora et al., 2017), the utilization of deep reinforcement learning for strategy games played in real time is discussed. The deep RTS gaming atmosphere is described in this study as a platform to investigate innovative AI methods in actual-time strategy games. An efficient game called Deep RTS was created exclusively for AI research. Reinforcement learning has already been used in easier game contexts such as that present in the Arcade Learning Conditions, but it has not been effectively utilized in more complex games. Researchers propose Deep RTS, a brand-new gaming platform geared toward deep reinforcement learning research, in this publication. The well-known Blizzard Entertainment video game StarCraft II served as the inspiration for Deep RTS, an RTS simulator. Research on strategy, deductive thinking, and handling at various stages of challenge are possible in the Deep RTS game scenario. This work draws influence

from microRTS and StarCraft II, with the objective of creating a setting that includes competitions among each of them. The goal of the Deep RTS attempt is to construct an outpost with an area hall and then work to extend the outpost with assets collected to obtain the military edge. Threats are carried out by military troops with the main objective of destroying the enemy's outpost. Initially, each player has a labor group. Expanding the outpost's harmful, protective, and resource-gathering capabilities is the main goal of the laboring groups in the game universe. Added groups that improve the player's attacking potential can arise from buildings. A player needs to eliminate each competitor group to achieve the finishing condition. Three levels can be used to describe a typical RTS game. The collecting and outpost development phase is the first level. The second level emphasizes financial and military power, whereas the third level is typically a death match among each team until the game is over. There are numerous game situations, including resource collection activities, military duties, and protecting tasks, which reduce the difficulty of a full real-time strategy game since deep RTS targets a diverse variety of reinforcement learning jobs. Deep RTS aims to realistically replicate RTS situations with super outstanding efficiency. The speed at which the game system changes the status of the game and the speed at which the game content may be produced as an image serve as performance indicators. A powerful RTS simulator, the Deep RTS game platform, offers quick investigation and testing of cutting-edge reinforcement learning methodologies.

In paper (Yu et al., 2018), DeDOL is one of the initial experiments for challenging long-form security games using deep Q learning. DeDOL (Deep-Q Network based Double Oracle enhanced with Local modes), a deep reinforcement learning-based procedure, is used to construct a monitoring approach for zero-sum Green Security Games (GSG) that responds to the actual time input. To simulate the operational relationship between criminal protection authorities, the Green Security Games (GSG) was developed (known as defenders) and its enemies (known as attackers) in areas of green security.

The defender and the attacker are the only two participants in the game. The attacker selects a single entrance point x from a set of entrances at the start of the interaction, while the defender always begins from the patrol station. The attacker's initial attack power is limited tools and uses them to launch attacks at the targets he has in mind. This game model is suitable for several green security areas. Until an ultimate time step T is achieved, the interaction concludes or when the attacker and all of the attack tools are discovered by the defender. The overall prize for the defender is the outcome of the game. In this study, they focus on zero-sum games, in which the attacker also receives rewards proportional to these actions. The game presumes that both participants have access to local observations. They just look at the current position of their competitor's footprints. Instead of using the entire grid, they only use one cell to represent the reality that they frequently have a restricted view of their surroundings because of the thick foliage, challenging terrain, or severe weather.

Deep Q-learning (Hasselt et al., 2015) is utilized to estimate the optimal output with the convolution neural network. The learning of the vanilla version of the deep Q-network (DQN) was challenging due to the GSG-I environment's high level of

dynamicity, particularly when the other player employed a randomized approach. Therefore, to increase learning stability and loss, it uses double DQN approaches. Finally, DeDOL is presented for zero-sum GSG-I, but it can be modified for general-sum GSG-I, particularly when the game is near zero sum.

Learning with opponent-learning awareness (LOLA) is an approach where every player in the surroundings influences the predicted training of the other competitors. The LOLA developing rules have an additional term that takes into consideration how one agent's policy may affect another agent's predicted attribute updates. The iterated prisoners' dilemma (IPD) leads to the establishment of tit-for-tat and, as a result, cooperation when two LOLA agents come into contact, but independent learning does not take place. In this domain, LOLA also benefits more than a naive learner and is resistant to being taken advantage of by higher-order gradient-based approaches. Recent years have seen a boom in multiagent reinforcement learning (RL) because of the development of RL techniques that enable the research of numerous agents in rich surroundings (Fukushima, 1980).

The learning outcomes in games with cooperative and competitive components have long been studied in game theory. Specifically, IPD is frequently used to study the conflict of cooperation and defection. In this game, pursuing selfish goals may result in poorer overall for all agents, but working together increases social welfare, a particular indicator of which is the total benefits received by all agents.

Another clause in the LOLA training rule takes into consideration how one agent's policy affects the other agents' next steps in learning. The strategy is not restricted to zero-sum games and could be used in general-sum situations as well. LOLA was applied to the deep RL setting utilizing likelihood ratio policy gradients, allowing it to be scaled to settings with high-dimensional input and parameter spaces. LOLA leads to high social welfare cooperation, whereas independent policy gradients, a conventional multiagent RL technique, do not follow. LOLA is applicable in situations when the opponent's policy is unknown and must be deduced from observations of the opponent's actions. In IMP, LOLA results in stable learning of the Nash equilibrium. In a round-robin contest against other multiagent learning algorithms, precise LOLA agents obtain the highest average returns on the IPD and reasonable performance on IMP.

In paper (Lanctot et al., 2017), we see a renewed interest in multiagent reinforcement learning (MARL) as a result of the achievement of deep reinforcement learning. In MARL, multiple agents communicate and gain insight in the same world at the same time, either competitively or cooperatively. Independent RL (InRL) is the most basic form of MARL, in which each agent is unaware of other agents and basically considers every interaction as part of its ("localized") surroundings. It presents a novel metric for measuring the correlation impact of strategies learned by independent learners, as well as demonstrating the degree of severity of the overfitting issue. It proposes a novel algorithm that employs Deep RL to identify the optimal outcomes from a policy distribution and a scientific game-theoretic study to calculate novel meta-strategy distributions. It assumes centralized learning for decentralized execution, as is typical in the MARL setting.

Policy-space response oracles (PSRO) are the primary conceptual algorithm. The algorithm is a natural generalization of Double Oracle in which the options in the meta-game are policies rather than actions. The procedure is documented on paper. The meta-game is depicted as an empirical game, beginning with a single policy (uniform random) and expanding each epoch by adding policies ("oracles") that approach best responses to the other players' meta-strategy. Once the other players in a partially observable multiagent environment remain constant, the environment becomes Markovian, and finding the optimal response simplifies to solving a form of MDP. In every episode, a single player is placed in oracle (learning) form to learn, and a fixed policy is chosen from among the opponents' meta-strategies. Although the generalization of PSRO is obvious and fascinating, the RL phase may require a long time to reach a satisfactory result. In challenging circumstances, many of the basic behaviors gained in one epoch may have to be relearned when restarting over. To address these issues, we propose a practical parallel version of PSRO. Instead of an infinite number of epochs, we predetermine a set number of levels. This is known as deep cognitive hierarchy (DCH).

It proposes joint policy correlation (JPC) matrices to determine the impact of overfitting in independent reinforcement learners. It demonstrates that PSRO or DCH generates general rules that decrease JPC drastically in partially observable cooperative games, as well as powerful counterstrategies that securely attack opponents in a usual competitive unreliable data game.

An approach for predicting driver behavior in scenarios involving highway driving is proposed, merging deep reinforcement learning and hierarchical game theory. Plenty of distance for driving tests is thought to be necessary for self-driving vehicles to achieve the same standard of assurance as cars with drivers. Driver models for game theory (Albaba & Yildiz, 2021) are contrasted with actual human driving patterns of behavior collected from traffic data to show the authenticity of the suggested modeling framework. In the open literature, a number of methods are suggested for obtaining highly accurate representations of human drivers. For the detection and forecasting of driving actions such as braking and steering control, Markov dynamic models (MDMs) are used. It is suggested to use a method called SITRAS (simulation of intelligent transport systems) to mimic lane change and merging behaviors. Dynamic Bayesian networks (DBNs) can recognize lane shifting or speeding. Therefore, a number of articles have been published regarding problems related to self-driving vehicles.

The paper offers a probabilistic modeling approach using game theory and deep reinforcement learning that enables concurrent choice making for multiagent traffic scenarios. By modeling the decision-making ego driver and supposing fixed responses for the other drivers, this approach differs from previous research because every driver in a multimove situation executes strategic choices concurrently. This is accomplished by fusing deep Q-learning (DQN), a reinforcement learning technique, with level-k reasoning, which comes from hierarchical game theory (Albaba & Yildiz, 2021). Level-k logic is a type of game theory concept that is utilized to simulate the strategic thinking process of human drivers. The level-K method is a hierarchical method of making decision framework that

assumes that multiple individuals have varying capacities for thinking. Level-k reasoning is insufficient on its own. Deep Q-learning (DQN) and level-k reasoning are combined to produce driver designs that respond well for each of the agent's predicted behaviors in a multimove scenario. In this study, the model is validated using two collection of traffic data that were gathered from the US101 and I80 motorways. The US101 set is chosen from these two sets to calculate the values of the observation and action space parameters. Two distinct reinforcement learning (RL) techniques, deep Q-learning (DQN) and its continuous counterpart, c-DQN, are utilized in conjunction with the level-k reasoning method to train driver policies. The Kolmogorov-Smirnov goodness of fit test (K-S test) is used to compare the suggested regulations, or driver algorithms, with the policies derived by analyzing real traffic data.

The authors (Xu et al., 2022), discussed the Internet of Vehicles (IoV). The IoV is a flexible portable networking framework that unites customers, automobiles, detecting equipment, and providers of services to enable connections between automobiles and people on the connection. Automobiles typically have limited computational capabilities, which prevents them from processing large volumes of service requests quickly. However, in the circumstance of an automobile running at a fast speed, time-dependent customer requests like collision alerts need to be handled right away. The quality of service (QoS) of automotive operations in the IoV is partially improved by cloud computing by reducing operation execution latency. The emergence of edge computing (EC) offers an additional advantageous approach to address the drawbacks of cloud computing in information transfer. By placing workstations in roadside units (RSU) that are adjacent to towns, EC as a fresh approach to computing dramatically minimizes the length of information delivery. In most cases, the edge device has constrained storage and computational capabilities. The customer's assistance execution plan can be identified through smart controlling methods such as reinforcement learning where the amounts of RSUs and requests for services are minimal.

It is difficult to maintain the QoS for the facilities in the IoV while sticking to the restriction of RSU assets, necessitating the creation of a fair service offloading mechanism according to the estimate of future traffic flow. To increase resource usage, the RSU have to maximize the distribution of hardware and software assets. At the moment, RSU load state and support selections for unloading must therefore be optimized using a reliable quick traffic flow prediction approach. In this research, a service offloading method based on game theory approach for cutting-edge technology in the IoV is suggested to address those issues. The Takagi-Sugeno fuzzy neural network (T-S FNN) is used in the technique to forecast immediate traffic flow. The T-S FNN uses a hybrid technique of back propagation and least squares algorithm to change parameters as opposed to the conventional traffic prediction method (such as ST-ResNet), which includes more hidden neurons and learning parameters. The T-S FNN is built on the Sugeno model, which integrates neural networks and fuzzy systems and uses them to modify variables in fuzzy control. A fuzzy neural network with multiple inputs and one output is created. The goal is to raise the system's QoS or to improve the overall QoS of all services provided to consumers.

The authors (Foahom Gouabou et al., 2022) introduced AI-assisted diagnosis, which is thought of as a potential remedy for early melanoma detection. Convolutional neural networks (CNNs) in particular have made significant breakthroughs in deep learning, although computer-aided diagnostic (CAD) technologies are not yet widely deployed in clinical settings. Game theory, a theory that concentrates on finding the best judgment alternative that enhances the judgment maker's expected benefits, is a potential remedy. A new paradigm for automatic melanoma detection is presented in this paper. An intriguing method for helping dermatologists with melanoma detection in real practice is the use of CAD systems. In fact, exceptional results by CAD driven by CNNs have been seen, matching dermatologists in a clinical trial. The majority of the frequently suggested methods for enhancing skin cancer detection accuracy fall into one of three groups: ensemble learning, data augmentation, and transfer learning.

The objective of this work is to build a reliable CAD for melanoma diagnosis and to explain the reasoning behind it. Models were created from pretrained designs and obtained training data enhanced with artificial photos acquired using artificial informational data creation methods because the workflow of the research is an ensemble method mixing multiple models. Considering the resources accessible, the models used in this work on the B5 version of EfficientNet. In place of the classification layer, the author modified the original system by introducing a second fully connected (FC) layer for binary operations that use two nodes for categorization or three nodes for ternary categorization.

To create a method of decision-making that is intelligible by users and nonexperts, the study offers a novel hierarchical architecture motivated by game theory. Theoretically, conflict between rival players can be modeled using game theory, which also allows for the analysis of different players' behavior. Two strategies are outlined for the players in the game theory method used in this work. The initial strategy is to group the guesses by degree of confidence unsure, and the subsequent action is to create the ultimate forecast. The output probabilities of the algorithms determine the payout value.

Moreover, a novel method for determining the level of confidence in an estimate has also been provided. The structure has been integrated with a heatmap display to enable a good understanding of outcomes. This strategy would enable our CAD to increase transparency in decisions and outperform earlier approaches. The findings of this work clearly demonstrate that it is feasible to develop automatic medical systems that utilize deep learning that are explicable without sacrificing diagnostic precision.

The authors (Khadarvali et al., 2022) discussed the smart grid, and the most recent technology to produce and distribute the right quantity of power is known as smart grids. Most modern smart grid applications require load frequency control (LFC), which is critical. It offers superior control of two-area systems. Whenever 2 (two) power systems are linked to one another, an LFC is applied to determine the stability of the connection. There could be variations in the power and frequency whenever there's a significant change, like a generator shift or load

shift. The frequency error must be zero to synchronize the two systems. To do this, fuzzy logic controls (FLC), proportional integral controllers (PIC), and integral controllers (IC) are all utilized. An artificial neural network (ANN) that was trained using the results of an ant lion optimization (ALO)-tuned PI controller (ALO-PI) was used by the author to replace the conventional integral controller in the two domains. Both the input and output information were obtained from the ALO PI controller. To improve control, a backpropagation technique was used to train the ANN.

The grid is becoming smarter, thanks to recent technological advancements, and wireless data communication is used. Wireless data transmission has a risk. Game theory has a significant impact on that. The frequency is one of the most important features to maintain, and this study focuses on a power system's most crucial component which is its stability. This is because a power system's frequency changes based on how much demand is being delivered to it, either increasing or decreasing. The frequency control system has three modes of operation. When a step-load fluctuation is introduced to the system, the primary and secondary frequency control in this situation constitute a parallel PIC that can drive frequency variations to zero. A control system called tertiary frequency control relies on offline enhancements. In this paper, different attacks in LFCs are mentioned, and the countermeasures of the mentioned attacks are presented. The traditional integral controller and the suggested ANN were compared, taking into account the behaviors of the attacker and defense, and the ANN surpassed the IC in the setting time.

The authors (Kishorea et al., 2020) discussed image restoration techniques. Restoration is a process in which it is necessary to boost the specific part of a picture that has distortions that are caused by damage or blur due to faulty image capture. Deep learning generative adversarial network (GAN) with two primary elements was proposed by the author: a discriminator (convolutional neural network) and a generator (deconvolutional neural network) model. In the generator model, the network is given input data and determines the input data's combined probability distribution to generate further points of data adopting the same distribution. With the discriminator model, the network is a simple model classifier that classifies the images produced by the generator as either different from or similar to the target image. The aim is to obtain images from the generator whose probability distribution is closest to that of the target image. The Nash equilibrium and minmax algorithm are two key ideas in game theory that relate to the battle between generators and discriminators. Gaining more error in the discriminative model is the primary goal of the generative network. A well-known dataset is used as training data for the discriminator. Use training dataset samples to achieve training with a high accuracy and low error rate. If the generator is able to trick the discriminators, then the generator will adjust its training.

Backpropagation is used in both neural networks, such that the generator is able to create a better result, and it also helps the discriminator by helping it recognize or

flag candidates for the synthesized images. In this article, they suggested a brand-new technique called EIRGAN. To deblur blurry images, the author introduces EIRGAN, an improved generative adversarial network for image restoration that is an alternative of Nash equilibrium. Additionally, they used the mathematical strategy outlined, restoring images using GAN training with the negative-f divergence function. By developing and evaluating the model on the same dataset as those other GAN designs, they have moreover contrasted our generative adversarial network with additional common and cutting-edge GAN designs that are accessible in the general public domain.

To study satellite behavior or to gather critical information from space, machine learning and game theory were used in paper (Shen et al., 2020). SSA stands for space situational awareness used for controlling satellite mobility, and it is dependent on quick and precise space object behavioral classification and finding. Due to the absence of a labelled dataset for the purpose of validation and training, it is the biggest barrier for using methods for machine learning (ML) and assessing their effectiveness in applications for SSA. They introduce a data augmentation methodology that incorporates game theory to generate datasets for ML methods for satellite behavior identification. Data such as elevation angle, azimuth angle, range, and range rate are transmitted through SGP4/SDP4. To create evasive maneuvering methods for space objects, they use a two-player game of pursuit evasion. RSO, which stands for resident space objects, is a technique used by them for resolving the issue related to SSA behavior detection, which is provided by the game theory approach. While the SSA observer improves tracking performance, RSO manipulates tracking estimates through a sensing and monitoring system to deceive the SSA observer. The analysis of RSO can be coordinated using UDOP, which stands for user-defined operation pictures, to correctly execute SSA. They also used GANs to further amplify the data that were simulated to enrich the training data.

3D-CNN is used to classify satellite behavior to assess performance. The filters of the convolutional network are selected randomly, resulting in random classification. Given the quantity of iterations increases, the accuracy of the training model reaches 100%. The rest of the dataset is utilized for the proficient CNN model's evaluation. The trained machine learning model has a 97% accuracy rate for classifying satellite behaviors. ResNet upgraded the GAN model so that it can correctly conduct data augmentation. The efficacy of machine learning algorithms during training and validation is enhanced by this model-based, game theoretic, synthetic data. The purpose of this study is to show how classification using deep learning models with the help of game theory activated, data augmentation, and produced training data can identify the behavior of space objects with greater accuracy.

In paper (Pal & Vidal, 2020), a game theory framework was suggested by scholars to study adversarial attacks and defenses. There is an existence of a set of attacks and defenses in the Nash equilibrium in their model given a linear hypothesis on the border of decision-making for the underpinning binary classification, which means

that there are some input modifications that the attacker and defender are permitted for, how much information does the attacker and defender know about one another, and then the approaches that are allowed to be specified. For deriving general constraints for the classifier for hypothetical test data, they provide an optimized approach to determine the equilibrium defense for any given classifier. Accessing a training set of n unique samples, researchers provide methods to approximate the optimal defense and derive generalized bounds on the finite-sample estimate's result. Their bounds demonstrate that the estimate progresses toward the best defense at a rapid rate of O(sqr (log n/n)) with respect to the sample count n. They concentrate on traditional game-theory techniques to the adversarial classification problem and other contemporary defenses for that can obtain theoretical assurances on the performance being attack. This classification study focuses on email spam detection, given a dataset (X, Y) and an adversary who can alter the positive (spam) by imposing a transformation cost determined by a cost function c. The defender is free to select a classifier h that divides any x belongs to X into spam and nonspam categories.

In contrast to previous research, defender in this study performs an additive perturbation rather than a classifier. They treat the base classifier as fixed and are given to us to attack or defend in accordance with practice. They also concentrate on the scenario when the attacker is capable of adding disturbances. This work seeks to describe adversarial attacks and defenses in a way that may be game theoretically optimal for each other, in which the attacker is unable to reduce reliable accuracy if defense is fixed and the defender cannot improve reliable accuracy if attack is fixed.

In paper (Wong et al., 2021), the aim is to analyze the allocation of resources issue in multicell MIMO networks and to develop an algorithm with distributed computation that can maximize the frequency to every base station user base stations with CSI at every base station. CSI stands for channel state information. MIMO stands for multiple-input multiple-output systems that are used for exceptional capacity using the quantity of antennas at the input and output ends. However, practically, the number of antennas should not be too high. For example, in the case of 5G, the number of antennas should not be greater than 64. The problem of interference can be solved by the channel state with the help of zero-forcing, as all users have the same frequency channels within the same cell. The next task is to optimize the distributed DFA for all the base stations. DFA stands for dynamic frequency allocation. For frequency resources with one another, the base stations build up as an artificial forward-looking game, the DFA problem in which they compete strategically. The issue with game theory in this application is that an unbiased player is self-centered and solely concerned with their personal benefit, which is determined by both of their own tactics and the reactions of their rivals. A participant ought to maximize their approach predicated on the end payoff rather than the immediate reward. The player will have to think about things that are happening in the future as well as in the present. This study aims to jointly optimize in multicell scenarios

by utilizing the cooperation between MIMO and DFA created by games with a forward-looking aspect.

The base stations are trained to perfect their game-theoretic reconciliation strategies by utilizing multiagent deep reinforcement learning (DRL) with centralized offline training so that it optimizes the network capacity. Finally, each base station has a trained neural network that is equipped with extensive knowledge about reconciling with other base stations so that it converges to a network-efficient equilibrium. They also used QMIX architecture in conjunction with TD3 (a double Q-learning variation) for optimizing network capacity.

The authors (Hu et al., 2022) discussed how image segmentation is performed with the help of deep learning and game theory. They introduced an improved technique of the Unet-Ore neural network model to address the issue of inadequate segmentation. This is caused by the fuzzily segmented ore in each picture. For segmenting the ore images, both game theory and deep learning are used. The proposed Unet-Ore neural network differs from the conventional neural network structure. There is improvement in feature extraction and generalization capacity by changing some neural network structures of the conventional one.

The authors (Ardekani et al., 2022) discussed track selection for self-driving cars in complicated situations by using a unique method that is based on memory neural networks and Nash equilibrium. To assess the recommended strategy in the racing game, consider the two vehicles: the ego and the opponent. The suggested approach for this is known as GT-LSTM. First, memory neural networks are used to understand and predict the way the competitor's car behaved. Second, agents employ matrices for payoffs in games to decide which course of action will result in the greatest payoff, and third, PID controllers smooth out lane changes and path following. To make the decision-making choice easier and more accurate, the merging of these domains is performed (game theory and neural network). This is done by training neural networks to determine how the opponent car behaves. The suggested approach outperforms the other two methods in regard to producing simulation outcomes, with a rate of success of 55% compared to 15% and 32% for game theory without knowing the other player's vehicle and payoff matrices and game theory compared to traditional neural networks, respectively. In addition, it's been demonstrated that in 90.2 percent of the cases, the recommended algorithm's output corresponds to Nash equilibrium.

The authors (Mamoudan et al., 2022) discuss the pricing strategy of perishable food items. That takes into account both the product's brand value and the prices of rival producers. Think of yourself as a customer; if you are going to buy some product from a supermarket, what should you prefer, product A, whose expiration date is nearer, or product B, whose expiration date is far. The answer is obvious you will prefer product B. There are many factors that involve demand for the products; it can be the company name value, the product's price, demand rate, etc. The prices of the food products are set in the factory, and then, the salesman cannot change the price of the product. Therefore, for predicting a reasonable price, this model will

help. Algorithms powered by deep learning and artificial intelligence can be utilized to foresee the price strategies of competing producers. At both the micro- and macro-levels, it is important to predict how much different products will cost. As salesmen cannot change the price of the product, if products are not sold before they expire, significant expenses can be added to the supply chain. This results in increased environmental damage. The application of game theory, which can result in positive interaction, is consequently one useful tactic for managing the green supply chain.

The pricing model is presented using a game theory method. This model involves the supplier, the vendor, and consumers. The supplier's brand value has been taken into account as this model's variable. The company name can have a large impact on consumer behavior. Consumers' decisions to select and purchase things might be influenced by the brand. Accurate food price forecasting paves the way for the adoption of consumer and producer protection measures. It is preferable for the anticipated model to have a high accuracy and error should be low.

The recommended framework, known as CNN-LSTM-GA, blends CNN, LSTM, and a genetic algorithm (GA). This network may gather detailed characteristics from several variables. It is appropriate to use the LSTM layer for modeling temporal data gathered from unpredictable patterns of a series of time components, and the CNN layer is capable of extracting features between various variables that affect food prices. The CNN-LSTM model's hyperparameters were then tuned utilizing GA to minimize any potential mistakes and optimize the model. Then, they compare the proposed CNN LSTM-GA technique with other deep learning models using validation metrics including the root mean square error (RMSE), mean absolute error (MAE), R-square (R2), and mean absolute error (MSE). The measures reflect the degree of prediction error, and a smaller value indicates a more effective model. One more use of this model is related to seasonal food items; even if they are not perishable, they will have a significant storage cost.

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Chapter 5
Case Studies and Different Applications



Game theory is a branch of mathematics that studies interactions and decision-making in situations where the outcomes of one individual's choices depend on the choices made by others. It has a wide range of applications across various fields, including economics, politics, biology, psychology, and more. Some notable applications of game theory are discussed in this chapter.

5.1 Auctions

Auction theory is a branch of economics and game theory that studies the design and analysis of auctions. It explores how auctions work as mechanisms for allocating goods, services, or resources to potential buyers and how bidders strategically interact in the auction process. The main goals of auction theory are to understand the properties of different auction formats, predict the outcomes of auctions, and design auctions. All auctions are mechanisms that have allocation and award fee rules.

The main elements of an auction consist of the following:

Players: These players are bidders or individuals which compete and participate to obtain the item. They have their own values for items.

Strategies: These are basically plans to increase the players' chance of winning. They generally make decisions on their values and considering other players' actions and strategies.

Revenue Maximization: For the auctioneer, revenue maximization is often a primary objective. By understanding bidder behavior and strategic interactions, the auctioneer can design auction formats that optimize the revenue generated from the auction.

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There are different types of auctions which are discussed below (Cintuglu et al., 2015):

1. English auctions: Bidders generally compete by increasing their bids until no one is willing to increase the price. The auctioneer announces the current highest bid, and the process iterates until only one bidder remains.
2. Dutch auction: The auctioneer starts with a high price and lowers the price until any bidder accepts it.
3. First-price auctions: Bidders submit their private bids without knowing the bids of other bidders, and the highest bidder wins the auction and pays the amount they bid. The winning bidder is required to pay the amount they bid to the seller in exchange for the item being auctioned. Generally, they have applications in real estates, in house or property auctions.
4. Vickrey auction: This is where bidders simultaneously submit their values and send to the seller. Highest bidder will pay the second highest bid and gets the object.

Game theory helps analyze how bidders strategize and bid in auctions. Basically, it helps in decision making and predicting out various strategies. It also helps in telling outcomes with respect to other players, overall increasing payoffs. Game theory investigates how bidders strategically determine their bids to maximize their utility or profits. Game theory can also detect frauds in auctions by analyzing patterns and behaviors of auction.

Pricing Pricing decision in game theory refers to the strategic choices made by competing firms in setting the prices of their products or services. Game theory provides a framework to analyze the interactions and strategies of firms in a competitive market, and each firm's pricing decision affects its own profits as well as the profits of other firms and organizations.

Price Competition Pricing Games— Game theory is used to model and analyze pricing games where competing firms set their prices strategically to maximize their profits. In a situation where there are two firms in a market, the firms may engage in price competition to gain a larger market share by observing each other's moves. There are various models involved in pricing games; some of them are discussed below.

Bertrand Model: The Bertrand model is a classic example of a price competition game which is named after French economist Joseph Bertrand, where two firms simultaneously set prices, assuming they have similar products at different prices without knowing the price set by the other firm and the lowest price wins the entire market (Tremblay & Tremblay, 2011). In a situation where one firm sets higher price and the other sets lowest price, so obviously the firm offering lower price would win. One significant issue is that the assumption of simultaneous decision-making might not hold in all markets.

Nash Equilibrium: Game theory helps to identify the Nash equilibrium in price competition games, where both firms set prices at their marginal cost, resulting in zero economic profits. This equilibrium demonstrates that firms have limited pricing power in perfectly competitive markets.

Cooperative Pricing Games: Game theory helps to analyze cooperative pricing games, where firms in a cartel collude to set prices collectively to maximize joint profits.

Cartel Stability: Game theory is used to study the stability and sustainability of cartels as well as the factors that can lead to cartel breakdowns due to noncooperative behavior.

Let us consider an example of two organizations that have two possible strategies. They can choose a price of $5 or $10. Two hundred amounts of quantity are demanded for $5 and 400 for $10.

Case 1: At $5, the product demanded 200 quantities, so the profit would be 5*100 = 500 for each of them.

Case 2: If both have $10, then they are dividing 400/2 = 200, so, 200*10, which is equal to 2000 for both organizations.

Case 3: (5,10) only 5*200 = 1000 for A will be the profit and 0 profit for B.

Case 4: (10,5) B will warn 200*5 = 1000 and there will be no profit for A.

In Table 5.1, the Nash equilibrium will be ($5, $5), where both organizations will obtain 500 as profit. However, if the organizations cooperate, they obtain ($10, $10) as a solution with a profit equal to 2000 for both organizations.

Machine learning techniques offer a data-driven approach to studying strategic interactions and decision-making in game theory. Machine learning can be used to model and predict player behavior in strategic games. By analyzing historical data of players' actions and outcomes, machine learning models can learn patterns and tendencies, aiding in understanding strategic decision-making. Machine learning can be used to predict the best strategies. This prediction can help to analyze potential outcomes. Machine learning can simulate and analyze interactions among multiple agents in complex scenarios. Also, certain algorithms for auctions, market pricing, and resource allocation are developed. Machine learning algorithms can help to identify potential coalitions and estimate the prices and payoff benefits the firms would get if they collaborate. In recent times, machine learning is extensively being used for multiagent systems (Hazra & Anjaria, 2022).

Table 5.1 Payoff matrix

Organization A/B	$ 5 for B	$ 10 for B
$ 5 for A	500, 500	1000,0
$ 10 for A	0,1000	2000, 2000

5.2 Game Theory in GAN

Game theory has various applications in GAN models which are used for generation of synthetic data. It consists of generator and discriminator which are trained in aggressive way. Goodfellow et al. discussed about minimizing worst-case error which is possible using adversarial training of GAN (Goodfellow et al., 2014). GANs can be formulated as a minimax game between the generator and discriminator. The generator tries to produce data that can deceive the discriminator, while the discriminator aims to distinguish between real and generated data accurately. This formulation is analogous to a two-player zero-sum game in game theory. GANs can be formulated as a two-player minimax game (Goodfellow et al., 2014), where the generator and discriminator are the players and their objective is to maximize their own utility (minimizing their own loss) while considering the other player's actions. Game theory-inspired regularization techniques are employed to stabilize GAN training and improve the performance of both the generator and discriminator. Berthelot et al. designed a new loss function used in the training algorithm (Berthelot et al., 2017). In Oliehoek et al. (2018) GAN is used in finite games with mixed strategies along with local Nash equilibrium being achieved.

Nash Equilibrium and GAN: The Nash equilibrium in a GAN is determined when both the generator and discriminator have found their optimal strategies, and neither player has an incentive to change their strategy given the other player's strategy. At this equilibrium point, the generator produces data that are indistinguishable from real data, and the discriminator cannot differentiate between real and generated data. Mathematically, in a GAN, the generator seeks to minimize its loss function, which (Berthelot et al., 2017) measures how well it can fool the discriminator. Additionally, the discriminator aims to minimize its loss function, which measures how well it can distinguish between real and generated data. The training process involves iteratively updating the generator and discriminator in a competitive manner until they reach a Nash equilibrium. In the Nash equilibrium of a GAN, the generator produces samples that are similar to real data, and the discriminator is unable to differentiate between real and generated data. Liu and Chawla (2010) discussed a Stackelberg zero-sum game with two players, which is used to make a loss function (Foerster et al., 2018). They trained a convolutional neural network (CNN) as the learner. The adversary is the leader L, and the learner is the follower F. A "geometry-aware GAN" could potentially refer to a type of generative adversarial network (GAN) that incorporates geometric information or considerations into its generative process, such as information about shapes, sizes, positions, and orientations (Huang et al., 2019). A geometry-aware GAN might incorporate an encoder-decoder architecture. It has various applications, such as converting 2D floor plans into 3D models or generating 3D models from 2D images. To generate scenes or environments, a geometry-aware GAN could be considered. A geometry-aware GAN could generate visualizations that accurately represent the spatial positioning of players, resources, and interactions, aiding in the analysis of game

dynamics. In multiagent settings, geometry-aware GANs might capture and generate player behaviors that respect geometric constraints. Since developments in both machine learning and game theory can emerge rapidly, there might be more recent research or applications related to geometry-aware GANs and game theory. It could be easily incorporated into and improve any existing GAN architecture (Kossaifi et al., 2017). GAN also uses supervised learning to estimate cost function and are also used for evaluation purposes (Hazra & Anjaria, 2022). Also, GANs could be used in continuous, non-convex games (Goodfellow 2016).

5.3 Game Theory in CNN

Game Theory and CNN: The application of convolutional neural networks (CNNs) in game theory is an emerging area of research that explores how CNNs can be used to analyze strategic interactions, predict player behavior, and model decision-making processes in various games. CNNs can be employed to analyze strategic interactions in various games, including board games, card games, and video games. By inputting game states or actions as images, CNNs can learn patterns and features that represent optimal strategies, winning moves, or potential threats. CNNs can be used to predict player behavior in games by analyzing their actions, game states, and historical data. This prediction can help game designers to understand players' preferences, adapt game content dynamically, and enhance the gaming experience. CNNs can provide strategy recommendations to players based on the current game state. By analyzing past game playing data, CNNs can suggest optimal moves or strategies to players, assisting them in decision-making during the game. CNNs can be employed to generate game content, such as level designs, game maps, and virtual environments. CNNs need modifying hyperparameters such as dropout rate, learning rate, and regularization parameters, which could be seen as a game theory scenario. Hyper-parameter arrangement is equivalent to a player's strategy, and the goal is to attain equilibrium. By learning from existing game content, CNNs can generate new content that adheres to the rules and constraints of the game (Hazra & Anjaria, 2022). CNNs can be used to represent game states in extensive-form games. By encoding the game tree structure and player actions into an image-like format, CNNs can learn to recognize patterns and features that represent the state of the game. CNNs can be trained to learn strategies from large datasets of game plays. This can provide insights into how players adapt their strategies over time, which is particularly useful in dynamic and repeated games. CNNs can predict the likely outcomes of extensive-form games based on players' moves and strategies. This prediction can help to assess the potential success of different approaches in a game setting. In wave net model, CNN is used to generate realistic musical waveforms. Also, GANs use CNN for image generation task, and the interaction between generator and discriminator resembles min-max game (Hazra & Anjaria, 2022).

5.4 Game Theory and Reinforcement Learning

Game Theory in Reinforcement Learning: In multiagent reinforcement learning, multiple agents interact in an environment and learn from their experiences to improve their strategies (Foerster et al., 2018). Game theory concepts, such as Nash equilibria and best responses, can be used to analyze the strategic behavior of agents in MARL settings. The agents' learning process can converge to Nash equilibria, which represent stable points of interaction in the game (Foerster et al., 2018). Some reinforcement learning scenarios involve adversarial interactions, where agents are competing against each other. Game theory can help to model these adversarial interactions and study the strategies and counterstrategies employed by agents. In reinforcement learning, agents learn from repeated interactions with their environment. This process is analogous to repeated games in game theory, where players learn from their past experiences to improve their strategies. Game theory has been used to analyze the interaction between attackers and defenders in the context of adversarial attacks on deep learning models. By viewing this scenario as a two-player game, researchers can develop robust defense strategies against adversarial attacks and improve the resilience of deep learning models.

5.5 Other Applications

An oligopoly could actually be considered as a market structure that consists of small firms that compete with each other (Tremblay & Tremblay, 2011). Any strategic move, such as changing prices or quantities, can affect rival firms, which makes decision-making complex. Oligopoly involves non-price competition, such as advertising, marketing, customer service, and innovation (Tremblay & Tremblay, 2011). Game theory helps to analyze the strategic decisions of firms in terms of choosing their pricing or production levels to maximize their profits while considering the reactions of their competitors. Oligopolies are often analyzed using game theory to model the strategic interactions among firms. Concepts such as Nash equilibrium, the prisoner's dilemma, and tit-for-tat strategies are commonly applied to understand the behavior of firms in oligopolistic markets. Oligopoly is characterized by a small number of interdependent competitors, which gives rise to complex decision-making processes and strategic interactions. In an oligopoly, there are few completed firms, and there is interdependence between the firms. These firms often engage in non-non-price competition. This includes advertising, product quality improvements, and customer service enhancements, which result in both mutual interests and conflicts among the firms. The study of oligopolies in game theory often involves using various models, such as the Cournot model, Bertrand model, and Stackelberg model, to analyze the behavior and decision-making of firms in these market structures.

TD-Gammon is a combination of neural networks and reinforcement learning that was developed by Gerald Tesauro in the late 1980s. The neural network was trained to predict the outcomes of moves and evaluate positions. The network weights were updated in each iteration. It reached a level of play that surpassed the best human players of the time. In (Sutton & Barto, 1960), TDG also uses a three-layered ANN architecture along with a reinforcement learning technique called TD-Lambda by Richard S. Sutton and Barto (1960). ANN basically predicts all the moves and selects the best move in iterations.

AlphaGo is a computer program developed by DeepMind. Go is an ancient Chinese board game with simple rules but an extremely large number of possible positions, making it vastly more complex than games such as chess. AlphaGo utilized deep neural networks to evaluate game positions and predict moves. These networks were trained on a large dataset of human games to learn patterns and strategies, combined neural networks with a Monte Carlo tree search algorithm (Silver et al., 2016), which is a heuristic search algorithm, and as the game developed, it was played against itself, and reinforcement learning was used and is known as "AlphaGO Zero" (Silver et al., 2017). Sometimes, CNNs are used to analyze the current board state, and recurrent neural networks (RNNs) help to capture the game's temporal dynamics. TorchCraft is a research framework that serves as a bridge between video games and machine learning platforms. It provides an interactive environment between the StarCraft game engine and AI frameworks, enabling AI agents to interact with the game environment. Basically, a reinforcement learning environment is provided (Synnaeve et al., 2016). TorchCraft has been used by researchers and AI enthusiasts to develop and share their AI-driven agents. It helps to make decisions in dynamic and strategic environments. Handling real-time situations and having imperfect information on actions and strategies is a challenge for the framework.

Generative moment matching networks are a type of model used for generative tasks, particularly for generating data that closely match the underlying data distribution (Li et al., 2015). Basically, matching of the mean and variance of the generated data distribution to the real data distribution is performed. They use neural networks as their primary modeling tool. They can be applied to tasks such as image generation, data augmentation, and feature learning. It helps in creating realistic simulation environments for testing different strategies and outcomes. Game theory often involves working with limited or incomplete data. GMMNs could aid in data augmentation by generating additional samples that capture the distribution of the available data, helping to improve the reliability of analysis and conclusions. In multiagent settings, it could also be used to mimic the behavior of other players by generating synthetic data.

Autoencoders are a type of neural network architecture used for unsupervised learning in machine learning (Vincent et al., 2010). They also help to reduce noise and help in reduction of dimensions (Hazra & Anjaria, 2022). In game theory, the payoffs and strategies for multiple players can lead to complex data structures. Autoencoders could help to reduce the dimensionality of the data. They can analyze equilibrium points in strategic games. By training an autoencoder on the observed actions or decisions of players, patterns and features of their behavior can be

captured. Generating new instances of game-related data, while maintaining the essential characteristics of the game, could improve the reliability of analysis, which is achieved by using autoencoders. Coevolutionary neural population models are a type of computational framework used to simulate the interactions and evolution of populations of neural networks (Moran & Pollack, 2018). Basically, they are used in solving tasks where the environment is dynamic and changes from time to time. In these models, there are multiple neural populations, each representing neural networks. Fitness functions are used to measure the quality of their solutions. They are used to solve optimization and adaptation problems. These models also have applications in fields such as artificial intelligence, neuroscience, and evolutionary computation.

Coevolutionary neural population models (Moran & Pollack, 2018) can simulate how agents adjust their strategies dynamically in response to the strategies employed by other agents. Coevolutionary models can represent the evolution of strategies over generations of play. In game theory, this could involve simulating the evolution of strategies in scenarios such as repeated games, where players learn from past interactions. Some games involve intricate and multistep strategies. Coevolutionary models might reveal how agents learn and develop such complex strategies through interactions.

Game Theory and Computer Vision: Game theory can be used to model and analyze strategic interactions among multiple agents in visual scenes. For example, in a surveillance scenario, multiple cameras might compete to observe certain areas effectively while avoiding detection by potential intruders. Computer vision techniques can be applied to track and identify objects or entities in visual data. Computer vision can be utilized in designing and testing new game mechanics, features, and user experiences, helping developers.

Game theory has several potential applications in the field of farming, particularly in regard to decision-making and resource allocation. Game theory can be used to model the allocation of water resources among competing agricultural users. Game theory can model the interactions between farmers and suppliers (e.g., seed, fertilizer) to analyze pricing and negotiation strategies. Game-theoretic models can explore how farmers' decisions about land use and crop selection affect local biodiversity, leading to more sustainable agricultural practices. Game theory can analyze how farmers' decisions about pricing, production, and timing affect their competitiveness in markets. It can also help farmers adapt to changing dynamic environmental conditions and climatic conditions, which are changing daily.

Game theory has many applications in educational institutions, where it can help in creating models of the interactions between students and colleges during the admission process, helping to understand strategies such as early decision applications and merit-based selections. Game theory can analyze how teachers design grading and incentive systems. Game theory can analyze how students collaborate and make decisions in group projects. Game theory can be applied to design adaptive learning platforms that provide educational content and experiences to individual student needs and learning styles.

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Chapter 6
Conclusion and Future Research Directions



In the concluding chapter of *Game Theory in Deep Learning*, we endeavor to encapsulate the book's essence, weaving together a comprehensive summary of its content with a deep dive into the pivotal insights gleaned from each section. This chapter serves as a thorough recapitulation of the book and carves a path forward for future research in this intriguing field. We commence by revisiting the significant findings and theories presented, distilling the wealth of information into key takeaways that highlight the profound intersection of game theory and deep learning. This synthesis is designed to foster a deeper understanding of the subject and to ignite a passion for further exploration and innovation in these dynamic and intersecting fields of study.

6.1 A Summary of Key Insights

The book *Game Theory in Deep Learning* explores game theory, a fascinating branch of mathematics that delves into strategic decision-making among multiple interacting players. The chapter lays a solid foundation by introducing the core principles of game theory and demonstrating its widespread application across various fields, including economics, political science, and evolutionary biology. It emphasizes game theory's pivotal role in understanding human behavior in strategic situations, its valuable contributions to business strategy, and the predictive analysis of events involving multiple players. The book then categorizes games into cooperative and noncooperative types, illuminating different aspects of strategic interaction. In cooperative games, players work together toward a common goal, whereas in noncooperative games, they pursue individual strategies and compete. A key concept introduced here is the Nash equilibrium, a critical notion in both game types that represents an optimal solution in strategic decision-making scenarios. This concept is instrumental in shaping our understanding of strategic interactions within these games.

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Transitioning into the realm of deep learning, the book describes it as a sophisticated subset of machine learning that involves training intricate neural networks on large datasets. It explains how deep learning emulates human brain functions by processing information across multiple layers. It also discusses its applications in fields like image and speech recognition and natural language processing, as well as its challenges, particularly in dealing with nonlinear problems and data scarcity. Lastly, the book highlights the intersection of game theory and deep learning, showcasing how game theory can profoundly enhance deep learning. It covers the application of game theory in modeling multiagent system interactions, optimizing deep learning model performance, and bolstering security against adversarial attacks. The chapter also underscores the integration of game-theoretic principles in designing neural networks for tasks such as computer vision and the role of game theory in adversarial training, illustrating its potential to improve the robustness and efficiency of deep learning models.

Building upon the foundational insights of game theory and deep learning, the book delves into practically implementing these concepts in real-world scenarios. A particularly intriguing area of application is in the field of autonomous systems. Here, game theory provides a framework for understanding the strategic interactions of autonomous agents, such as drones or self-driving cars. For instance, in multiagent environments, these systems must make decisions that ensure their objectives but also account for the actions of others in the shared environment. The present book is aligned with the current research that illustrates the use of game theory in predicting the behaviors of autonomous agents and optimizing their decision-making processes.

Furthermore, the book explores the use of game theory in augmenting the capabilities of deep neural networks in complex decision-making tasks. For example, deep learning models can benefit from game-theoretic approaches in medical diagnosis and treatment planning to make more informed and strategic decisions. This is particularly relevant when multiple outcomes must be weighed and optimized. The book provides a theoretical understanding of game theory and deep learning and offers a window into their practical applications and future research potential. It guides academics and practitioners, inspiring further exploration and innovation in these interrelated fields. The following subsection discusses the open questions and challenges regarding applying game theory in deep learning.

6.2 Open Questions, Challenges, and Cross-Disciplinary Opportunities

The application of game theory to deep learning, a rapidly evolving area of artificial intelligence, has shown promising results and potential in various research fields. However, this integration also presents several open questions and challenges. The present subsection discusses some open questions and challenges in applying game theory in deep learning.

The first challenge is about model complexity (Rodriguez et al., 2022). The complexity of combining game-theoretic models with deep learning architectures can be challenging. This includes the difficulty of ensuring the models are accurate and efficient while retaining the interpretability of their decisions. The performance and relevance of deep learning models, mainly when applied in a game-theoretic framework, are highly contingent on both the availability and the quality of data. This reliance becomes problematic in scenarios characterized by limited or substandard data, as it can significantly impair the model's effectiveness and practical applicability (Hu et al., 2021). The challenge lies in obtaining and curating high-quality datasets that faithfully inform and train these complex models. Within multiagent systems, the task of accurately modeling strategic interactions using game theory in conjunction with deep learning frameworks is notably intricate. It encompasses the ability to predict behaviors and outcomes in dynamic and frequently unpredictable environments. This complexity is heightened by the need to account for many variables and potential interactions, making it a particularly challenging aspect of research in this field. While game theory has been instrumental in bolstering the defense mechanisms of deep learning models against adversarial attacks, developing systems robust enough to counter sophisticated and evolving cyberthreats remain. This necessitates continuously refining defensive strategies to ensure these models remain impervious to malicious attacks.

In the context of game-theoretic models, especially when dealing with expansive and intricate deep learning systems, identifying optimal strategies and achieving equilibrium states present significant computational and algorithmic hurdles. The complexity of these systems often entails a substantial demand for computational resources and sophisticated algorithms to navigate the search for equilibrium solutions efficiently (Lins et al., 2021). The effective amalgamation of game theory and deep learning insights necessitates a profound comprehension of both domains. Successfully bridging these two fields cohesively and practically poses a formidable challenge for researchers, calling for an interdisciplinary approach that harmonizes each discipline's methodologies and theoretical underpinnings.

Furthermore, the ethical considerations and real-world applicability of models that combine game theory and deep learning cannot be overlooked (Anjaria, 2021). This is particularly critical in sensitive domains such as healthcare, finance, and the development of autonomous systems, where the consequences of decisions made by these models can have significant real-world impacts (Wu, 2022). Ensuring that these models adhere to ethical guidelines and are designed with practical utility remains a paramount concern. Addressing these challenges requires a multifaceted approach, blending advanced technical solutions with thoughtful consideration of ethical and practical aspects. As the field evolves, these challenges present obstacles and opportunities for groundbreaking research and development.

As we navigate these multifaceted challenges, the future of integrating game theory with deep learning shines with potential. Innovative approaches are needed to simplify model complexity, such as developing new frameworks that balance accuracy, efficiency, and interpretability (Kim et al., 2022). Embracing advancements in data science could alleviate data scarcity and quality issues, leveraging

techniques like synthetic data generation and advanced data augmentation. In strategic interaction modeling, the focus should shift toward more adaptive and resilient models that can handle the unpredictability inherent in multiagent systems (Ma et al., 2023). Further, enhancing adversarial robustness will require advanced algorithms and a continuous monitoring and updating mechanism to adapt to evolving cyberthreats. In terms of optimization and equilibrium finding, implementing more powerful computational techniques, perhaps drawing from emerging fields like quantum computing, could provide the necessary processing capabilities.

Interdisciplinary collaboration is critical to bridging game theory and deep learning effectively. Encouraging cross-disciplinary research initiatives and educational programs can foster a new generation of researchers with a holistic understanding of both fields. Finally, the ethical and practical aspects demand a comprehensive framework considering these technologies' societal, ethical, and real-world implications. This includes developing guidelines and standards for ethical AI and tailoring solutions to meet the specific needs of various sectors like healthcare and finance. Addressing these challenges and exploring these avenues can lead to groundbreaking advancements, making this integration a robust field of academic inquiry and a catalyst for innovative applications in artificial intelligence. The following section discusses the future research direction of applying game theory in deep learning.

6.3 Future Research Direction

The future of applying game theory in deep learning is ripe with exciting possibilities, encompassing diverse research avenues. One key direction is the development of hybrid models that merge the predictive prowess of deep learning with the strategic analytical strength of game theory. Additionally, there is a push toward crafting adaptive algorithms capable of navigating dynamic environments, especially critical in scenarios requiring real-time decision-making. Another pivotal area is enhancing the robustness of deep learning models against adversarial attacks through game-theoretic methods. The exploration of quantum computing's role in unravelling complex game-theoretic challenges within deep learning systems promises groundbreaking advances.

Moreover, the focus on ethical and fair AI development underlines the importance of creating morally sound and equitable models. Interdisciplinary studies are encouraged to amalgamate diverse insights, broadening the scope of game theory's application in deep learning. Real-world applications in sectors like finance, healthcare, and autonomous systems, where strategic decisions are crucial, are also a significant focus. Another exciting area is developing systems capable of automated strategy learning based on game-theoretic principles. Lastly, efforts are being directed toward formulating explainable AI models, integrating game theory and deep learning to make AI decisions more transparent and understandable to users. Each avenue addresses existing challenges and paves the way for innovative breakthroughs.

Future research should also focus on fine-tuning the balance between computational efficiency and the complexity inherent in these hybrid models (Kamal & Bablu, 2022). Particular emphasis must be placed on developing user-friendly interfaces that allow practitioners from various domains to leverage these models effectively. Additionally, establishing comprehensive benchmarks and standards for evaluating the performance of game theory-enhanced deep learning systems will be crucial. As the field progresses, fostering a collaborative ecosystem that brings together researchers, industry experts, and policymakers will be key to translating these advanced theoretical concepts into tangible, real-world solutions that can revolutionize various sectors.

In this evolving landscape, an essential aspect will be continuously exploring new datasets and environments to test and refine these game theory and deep learning models. This exploration will not only validate their efficacy across different scenarios but also uncover new challenges and opportunities for improvement. Integrating advanced technologies like augmented and virtual reality could also offer innovative platforms for simulating complex game-theoretic scenarios, enhancing the training and capabilities of deep learning systems (Zhu et al., 2022). Ultimately, the goal is to create a dynamic, iterative process of learning and adaptation, fostering an environment where both game theory and deep learning can evolve in tandem, driving forward the frontiers of artificial intelligence.

The future of applying game theory in deep learning holds immense promise, with several key research directions emerging. A primary focus is on developing new game-theoretic models that are more suited for deep learning applications. This entails designing efficient algorithms specifically tailored for solving game-theoretic problems within deep learning contexts. Another significant area is the integration of game theory principles directly into deep learning frameworks and tools, enhancing their strategic decision-making capabilities. Furthermore, evaluating the performance of these game-theoretic deep learning methods across various real-world problems will be crucial. As these fields continue to advance, we can anticipate a surge of innovative applications that leverage the strengths of both game theory and deep learning, making substantial contributions across diverse domains. These are just a few examples of promising future research directions for applying game theory in deep learning. As both fields continue to develop, we can expect to see even more creative and innovative applications of game theory to deep learning problems.

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